Eigenvectors of graph Laplacians: a landscape

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Abstract

We review the properties of eigenvectors for the graph Laplacian matrix, aiming at predicting a specific eigenvalue/vector from the geometry of the graph. After considering classical graphs for which the spectrum is known, we focus on eigenvectors that have zero components and extend the pioneering results of Merris (1998) on graph transformations that preserve a given eigenvalue λ or shift it in a simple way. These transformations enable us to obtain eigenvalues/vectors combinatorially instead of numerically; in particular we show that graphs having eigenvalues $\lambda = 1, 2, \ldots, 6$ up to six vertices can be obtained from a short list of graphs. For the converse problem of a λ subgraph G of a λ graph G", we prove results and conjecture that G and G" are connected by two of the simple transformations described above.

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1 Introduction

The graph Laplacian is an important operator for both theoretical reasons and applications [1]. As its continuous counterpart, it arises naturally from conservation laws and has many applications in physics and engineering. The graph Laplacian has real eigenvalues and eigenvectors can be chosen orthogonal. This gives rise to a Fourier like description of evolution problems on graphs; an example is the graph wave equation, a natural model for weak miscible flows on a network, see the articles [2], [3]. This simple formalism proved very useful for modeling the electrical grid [4] or describing an epidemic on a geographical network [5]. Finally, a different application of graph Laplacians is spectral clustering in data science, see the review [6].

Almost sixty years ago, Mark Kac [7] asked the question : can one Hear the Shape of a Drum? Otherwise said, does the spectrum of the Laplacian characterize the graph completely ? We know now that there are isospectral graphs so that there is no unique characterization. However, one can ask a simpler question: can one predict eigenvalues or eigenvectors from the geometry of the graph? From the literature, this seems very difficult, most of the results are inequalities, see for example the beautiful review by Mohar [8] and the extensive monograph [9].

Many of the results shown by Mohar [8] are inequalities on λ_2, the first non zero eigenvalue. This eigenvalue is related to the important maximum cut problem in graph theory and also others. Mohar [8] also gives some inequalities on λ_n, the maximum eigenvalue, in terms of the maximum of the sum of two degrees. Another important inequality concerns the interlacing of the spectra of two graphs with same vertices, differing only by an edge. However, little is known about the bulk of the spectrum, i.e. the eigenvalues between λ_2 and

λ_n. A very important step in that direction was Merris's pioneering article [10] where he introduced "Laplacian eigenvector principles" that allow to predict how the spectrum of a graph is affected by contracting, adding or deleting edges and/or of coalescing vertices. Also, Das [11] showed that connecting an additional vertex to all vertices of a graph increases all eigenvalues (except 0) by one.

Following these studies, in [12] we characterized graphs which possess eigenvectors of components ± 1 (bivalent) and $0, \pm 1$ (trivalent). This is novel because we give exact results, not inequalities. Here, we continue on this direction and focus on eigenvectors that have some zero coordinates, we term these soft nodes; such soft nodes are important because there, no action can be effected on the associated mechanical system [3]. In this article, we use the important properties of graphs with soft nodes, we call these soft-graphs, to highlight eigenvalues/eigenvectors that can be obtained combinatorially (instead of numerically). We first show that eigenvalues of graph Laplacians with weights one are integers or irrationals. Then we present well known classical graphs whose spectrum is known exactly. We describe five graph transformations that preserve a given eigenvalue and two that shift the eigenvalue in a simple way. Among the transformations that preserve an eigenvalue, the link was explicitly introduced in the remarkable article by Merris (*link principle*) [10]. The articulation and the soldering were contained in the same paper and we choose to present elementary versions of these transformations. We find two new transformations that preserve an eigenvalue: the regular expansion and the replacement of a coupling by a square. We also present transformations that shift an eigenvalue in a predictable way: insertion of a soft node, addition of a soft node, insertion of a matching. The first is new, the second and third were found by Das [11] and Merris [10] respectively.

In the last part of the article we enumerate all the small graphs up to six vertices that have a given eigenvalue λ and explain the relations between them using the transformations discussed previously. It is remarkable that these graphs can all be obtained from a short list of graphs. However the question is open for bigger graphs. Using the transformations mentioned above, λ soft graphs can be made arbitrarily large. The converse problem of a λ subgraph G of a λ graph G" is considered. We show that the matrix coupling the two Laplacians $L(G)$ and $L(G')$, where $G' = G" - G$, is a graph Laplacian. If the remainder graph G' is λ, then it is formed using the articulation or link transformation. It is possible that the remainder graph G' is not λ as long as it shares an eigenvector with G. Then the two may be related by adding one or several soft nodes to G'. Finally, an argument shows that if G' is not λ and does not share an eigenvector with G, the problem has no solution. We finish the article by examining the λ soft graphs for $\lambda = 1, 2, \ldots, 6$ and insist on minimal λ soft graphs as generators of these families, using the transformations above.
The article is organized as follows. Section 2 introduces the main definitions. In section 3 we consider special graphs (chains, cycles, cliques, bipartite graphs) whose Laplacian spectrum is well known. The graph transformations preserving

an eigenvalue are presented in section 4. Section 5 introduces graph transformations which shift eigenvalues. Finally section 6 introduces λ soft graphs, discusses λ sub-graphs and presents a classification of graphs up to six vertices.

2 The graph Laplacian : notation, definitions and properties

We consider a graph $G(V, E)$ with a vertex set V of cardinality n and edge set E of cardinal m where n, m are finite. The graph is assumed connected with no loops and no multiple edges. The graph Laplacian matrix [9] is the (n, n) matrix $L(G)$ or L such that

$$L_{ij} = -1 \text{ if edge i j exists}, 0 \text{ otherwise}, \quad L_{ii} = m_i, \text{ degree of i}, \tag{1}$$

where the degree of i is the number of edges connected to vertex i.

The matrix L is symmetric so that it has real eigenvalues and we can always find a basis of orthogonal eigenvectors. Specifically we arrange the eigenvalues λ_i as

$$\lambda_1 = 0 \leq \lambda_2 \leq \cdots \leq \lambda_n. \tag{2}$$

We label the associated eigenvectors $v^1, v^2, \ldots, v^n$.

We have the following properties

- $v^1 = \mathbf{1}$ the vector whose all components are 1.
- Let v^i_k be the k component of an eigenvector v^i, $\quad i > 1$. An immediate consequence of the v^i being orthogonal to v^1 is $\sum_k v^i_k = 0$.

A number of the results we present hold when $L_{ij} \neq -1$ and $L_{ii} = \sum_{j \sim i} L_{ij}$, this is the generalized Laplacian. We will indicate which as we present them.

Regular graphs
The graph Laplacian can be written as

$$L = D - A$$

where A is the adjacency matrix and D is the diagonal matrix of the degrees. We recall the definition of a regular graph.

Definition 2.1 (Regular graph) *A graph is d-regular if every vertex has the same degree d.*

For regular graphs $D = d\mathrm{Id}_n$, where Id_n is the identity matrix of order n. For these graphs, all the properties obtained for L in the present article carry over to A.

We will use the following definitions.

Definition 2.2 (Soft node) *A vertex s of a graph is a soft node for an eigenvalue λ of the graph Laplacian if there exists an eigenvector x for this eigenvalue such that $x_s = 0$.*

An important result due to Merris [10] is

Theorem 2.3 *Let G be a graph with n vertices. If $0 \neq \lambda < n$ is an eigenvalue of $L(G)$ then any eigenvector affording λ has component 0 on every vertex of degree $n-1$.*

Definition 2.4 (k-partite graph) *A k-partite graph is a graph whose vertices can be partitioned into k different independent sets so that no two vertices within the same set are adjacent.*

Definition 2.5 (cycle) *A cycle is a connected graph where all vertices have degree 2.*

Definition 2.6 (chain) *A chain is a connected graph where two vertices have degree 1 and the other vertices have degree 2.*

Definition 2.7 (clique) *A clique or complete graph K_n is a simple graph where every two vertices are connected.*

In the article we sometimes call configuration a vertex valued graph where the values correspond to an eigenvector of the graph Laplacian.

2.1 Eigenvalues are integers or irrationals

We have the following result

Theorem 2.8 *If the eigenvalue λ is an integer, then there exist integer eigenvectors.*

To see this consider the linear system

$$(L - \lambda I)X = 0.$$

It can be solved using Gauss's elimination. This involves algebraic manipulations so that the result X is rational. If X is rational, then multiplying by the product of the denominators of the entries, we obtain an eigenvector with integer entries.

We now show that the eigenvalues of a graph Laplacian are either integers or irrationals. We have the following rational root lemma on the roots of polynomials with integer coefficients, see for example [13]

Lemma 2.9 *Rational root*
Consider the polynomial equation

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 = 0$$

where the coefficients a_i are integers. Then, any rational solution $x = p/q$, where p, q are relatively prime is such that p divides a_0 and q divides a_n .

A consequence of this is

Theorem 2.10 *The eigenvalues of a graph Laplacian are either integers or irrationals.*

Proof. Consider the equation associated to the characteristic polynomial associated to the graph Laplacian, it has the form

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x,$$

because the graph is connected so that there is only one 0 eigenvalue. Assume that the eigenvalue is of the form $x = p/q$ with p, q are relatively prime integers. Then from the lemma above, p divides a_0 and q divides a_n. Since $a_n = \pm 1$, $q = 1$ so that $x = p$ is an integer. □

The fact that some graphs have integer spectrum was discussed by Grone and Merris [14]. Many of their results are inequalities for λ_2 and λ_{n-1}. Our results complement their approach.

3 Special graphs

3.1 Cliques and stars

The clique K_n has eigenvalue n with multiplicity $n-1$ and eigenvalue 0. The eigenvectors for eigenvalue n can be chosen as $v^k = e^1 - e^k, \quad k = 2, \ldots, n$. To see this note that

$$L = nI_n - \mathbf{1},$$

where I_n is the identity matrix of order n and $\mathbf{1}$ is the (n, n) matrix where all elements are 1.

A star of n vertices S_n is a tree such that one vertex , say vertex 1, is connected to all the others. For a star S_n, the eigenvalues and eigenvectors are

- $\lambda = 1$ multiplicity $n-2$, eigenvector $e^2 - e^k, \quad k = 3, \ldots, n$
- $\lambda = n$ multiplicity 1 , eigenvector $(n+1)e^1 - \sum_{k=2}^{n} e^k$
- $\lambda = 0$ multiplicity 1 , eigenvector $\hat{1}$

3.2 Bipartite and multipartite graphs

Consider a bipartite graph K_{n_1,n_2}. The Laplacian is

$$L = \begin{pmatrix} n_2 & 0 & \dots & 0 & -1 & \dots & & -1 \\ 0 & n_2 & 0 & \dots & -1 & \dots & & -1 \\ \dots & \dots & \dots & \dots & \dots & \dots & & \\ 0 & \dots & 0 & n_2 & -1 & \dots & & -1 \\ -1 & \dots & & -1 & n_1 & 0 & \dots & 0 \\ -1 & \dots & & -1 & 0 & n_1 & \dots & 0 \\ \dots & & & & & \dots & \dots & \dots \\ -1 & \dots & & -1 & 0 & 0 & \dots & n_1 \end{pmatrix}, \tag{3}$$

where the top left bloc has size $n_1 \times n_1$, and the bottom right bloc $n_2 \times n_2$. The eigenvalues with their multiplicities denoted as exponents are

$$0^1, \quad n_1^{n_2-1}, \quad n_2^{n_1-1}, \quad (n1+n2)^1.$$

Eigenvectors for n_1 can be chosen as $e^{n_1+1} - e^i \quad (i = n_1+2, \dots, n_1+n_2)$. The eigenvector for $n = n_1 + n_2$ is $(1/n_1, \dots, 1/n_1, -1/n_2, \dots, -1/n_2)^T$.

Similarly, the spectrum of a multipartite graph $K_{n_1,n_2,\dots n_p}$ is

$$0^1, \quad (n-n_1)^{n_1-1}, \quad (n-n_2)^{n_2-1}, \dots, \quad (n-n_p)^{n_p-1}, \quad n^p.$$

The eigenvectors associated to $n - n_1$ are composed of 1 and -1 in two vertices of part 1 padded with zeros for the rest.

3.3 Cycles

For a cycle, the Laplacian is a circulant matrix, therefore its spectrum is well-known. The eigenvalues are

$$\mu_k = 4\sin^2\left[\frac{(k-1)\pi}{n}\right], \quad k = 1, \dots, n. \tag{4}$$

They are associated to the complex eigenvectors v^k whose components are

$$v_j^k = \exp\left[\frac{i(j-1)(k-1)2\pi}{n}\right] \quad , j = 1, \dots n. \tag{5}$$

The real eigenvectors $w^k,\ x^k$ are,

$$w^k = (0,\ \sin(a_k),\ \sin(2a_k),\ \dots,\ \sin((n-1)a_k))^T, \tag{6}$$

$$x^k = (1,\ \cos(a_k),\ \cos(2a_k),\ \dots,\ \cos((n-1)a_k))^T, \tag{7}$$

$$a_k = \frac{2(k-1)\pi}{n} \tag{8}$$

Ordering the eigenvalues, we have

$$\lambda_1 = \mu_1 = 0, \tag{9}$$
$$\lambda_2 = \lambda_3 = \mu_2, \tag{10}$$
$$\lambda_{2k} = \lambda_{2k+1} = \mu_{k+1}, \tag{11}$$
$$\dots \tag{12}$$

For $n = 2p + 1$

$$\lambda_{2p} = \lambda_{2p+1} = \mu_{p+1}$$

For $n = 2p$

$$\lambda_{2p} = \mu_p = 4$$

is an eigenvalue of multiplicity 1; an eigenvector is $(1, -1, \dots, 1, -1)^T$. In all other cases, the eigenvalues have multiplicity two so that all vertices are soft nodes. Results for small n cycles are presented in table 1, where we indicated the existence of soft nodes. We omitted the zero eigenvalue mode $\lambda_1 = 0$.

n	eigenvalue	eigenvector	existence of soft nodes
3	$\lambda_2 = \lambda_3 = 3$	$(0, \sin\frac{2\pi}{3}, \sin\frac{4\pi}{3})$ $(1, \cos\frac{2\pi}{3}, \cos\frac{4\pi}{3})$	1 arbitrary soft node
4	$\lambda_2 = \lambda_3 = 2$	$(0, 1, 0, -1)$ $(1, 0, -1, 0)$	2 soft nodes with one arbitrary
5	$\lambda_2 = \lambda_5 = 4\sin^2\frac{\pi}{5} \approx -1.38$ $\lambda_3 = \lambda_4 = 4\sin^2\frac{2\pi}{5} \approx -3.62$		1 arbitrary soft node 1 arbitrary soft node
6	$\lambda_2 = \lambda_6 = 1$ $\lambda_3 = \lambda_5 = 3$		2 soft nodes with one arbitrary 2 soft nodes with one arbitrary
7	$\lambda_2 = \lambda_7 = 4\sin^2\frac{\pi}{7}$ $\lambda_3 = \lambda_6 = 4\sin^2\frac{2\pi}{7}$ $\lambda_4 = \lambda_5 = 4\sin^2\frac{3\pi}{7}$		1 arbitrary soft node 1 arbitrary soft node 1 arbitrary soft node

Table 1: *Eigenvalues and eigenvectors having soft nodes for small n cycles.*

Remark that the maximum number of 0s is $n/2$. To see this, note that if two adjacent vertices have value 0 then their neighbors in the cycle must have 0 as well and we only have 0s , but the null vector is not an eigenvector. This means that we have at most $n/2$ 0s. This bound is reached for n even.

3.4 Chains

For chains C_n, there are only single eigenvalues, they are [15]

$$\lambda_k = 4\sin^2(\frac{\pi(k-1)}{2n}) \quad , k = 1, \dots, n. \tag{13}$$

The eigenvector v^k has components

$$v_j^k = \cos\left(\frac{\pi(k-1)}{n}(j-\frac{1}{2})\right) \quad , j = 1, \ldots n. \tag{14}$$

Obviously the cosine is zero if and only if:

$$(k-1)(2j-1) = n(1+2m), \tag{15}$$

where m is an integer. There is no solution for $n = 2^\alpha$, for α a positive integer. Apart from this case, there is always at least one soft node. If n is a prime number, the middle vertex $j = (n+1)/2$ is the only soft node. For k odd, all vertices j such that $2j-1$ divides n have a zero value, including the middle vertex.

For n odd, chains and cycles share $(n-1)/2$ eigenvalues and eigenvectors. To see this consider a chain with $n = 2p+1$. All $k = 2q+1$ give a chain eigenvalue $\lambda_k = 4\sin^2(\frac{\pi q}{2p+1})$ that is also a cycle eigenvalue. The eigenvector components v_j^q are such that $v_1^q = v_{2p+1}^q$.

4 Transformations preserving eigenvalues

In this section, we present four main transformations of graphs such that one eigenvalue is preserved. These are the link between two vertices, the articulation, the soldering and the contraction/expansion. The first three transformations are in the literature in a general form; we choose to present them in their most elementary form.
Furthermore, these transformations will all be unary, they act on a single graph. Binary transformations can be reduced to unary transformations for non connected graphs.

Using these transformations we can generate new graphs that have a soft node, starting from minimal graphs having soft nodes.

4.1 Link between two equal vertices

An important theorem due to Merris [10] connects equal component vertices.

Theorem 4.1 Link *between two vertices : Let λ be an eigenvalue of $L(G)$ for an eigenvector x. If $x_i = x_j$ then λ is an eigenvalue of $L(G')$ for x where the graph G' is obtained from G by deleting or adding the edge $e = ij$.*

This transformation preserves the eigenvalue and eigenvector. It applies to multiple graphs. Fig. 1 shows examples of the transformation.

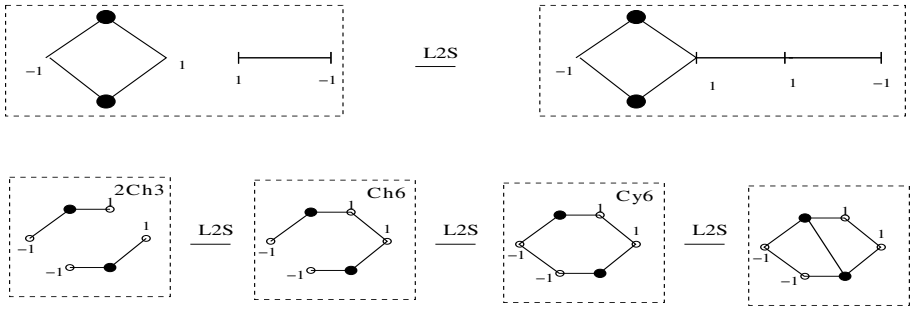


Figure 1: Example of the transform : link between two equal vertices.

We have the following corollary of the theorem.

Theorem 4.2 *Let λ be an eigenvalue of two graphs G_1 and G_2 for respective eigenvectors x^1, x^2 with two vertices i, j, such that $x^1_i \neq 0$ or $x^2_j \neq 0$. Then the graph $G(V_1 \cup V_2, E_1 \cup E_2 \cup ij)$ affords the eigenvector $y = x^2_j \begin{pmatrix} x^1 \\ 0 \end{pmatrix} + x^1_i \begin{pmatrix} 0 \\ x^2 \end{pmatrix}$ for λ.*

This allows to generate many more graphs that have an eigenvalue λ.

4.2 Articulation

An elementary transformation inspired by Merris's principle of reduction and extension [10] is to add a soft node to an existing soft node. This does not change the eigenvalue. We have the following result.

Theorem 4.3 Articulation (A) *: Assume a graph $G(V, E)$ with n vertices where x is an eigenvector such that $x_i = 0$ for an eigenvalue λ. Then, the extension x' of x such that $x'_{1:n} = x_{1:n}$ and $x'_{n+1} = 0$ is an eigenvector for λ for the Laplacian $L(G')$ where $G'(V', E')$ such that $V' = V \cup (n + 1)$ and $E' = E \cup i(n + 1)$.*

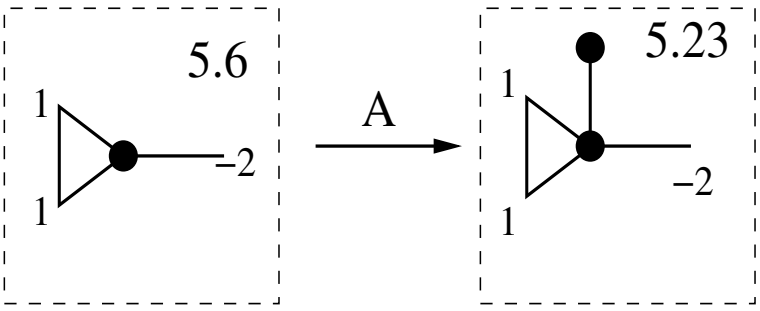


Figure 2: Example of the articulation property. The large dot corresponds to a soft node.

The general case presented by Merris [10] amounts to applying several times this elementary transformation.
The transformation is valid for graphs with arbitrary weights and the extended edges can have arbitrary weights.

Fig. 2 illustrates this property on the two graphs labeled 5.6 and 5.23 in the classification given in [1]. An immediate consequence of this elementary transform is that any soft node can be extended into an arbitrarily large graph of soft nodes while preserving the eigenvalue and extending the eigenvector in a trivial way. Fig. 3 shows two graphs that have the same eigenvalue $\lambda = 1$ and that are connected by the articulation transform.

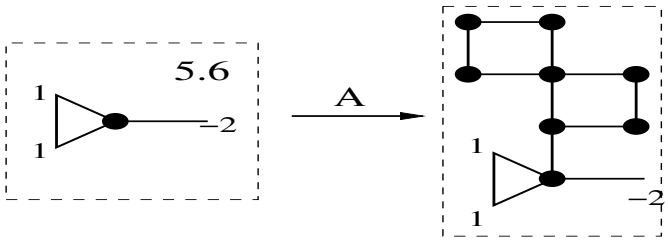


Figure 3: Two graphs connected by the articulation transform.

4.3 Soldering

A consequence of the contraction principle of Merris [10] is that coalescing two soft nodes of a graph leaves invariant the eigenvalue. This is especially important because we can "solder" two graphs at a soft node.

Theorem 4.4 Soldering *: Let x be an eigenvector affording λ for a graph G. Let i and j be two soft nodes without common neighbors. Let G' be the graph obtained from G by contracting i and j and x' be the vector obtained from x by deleting its jth component. Then x' is an eigenvector of $L(G')$ for λ.*

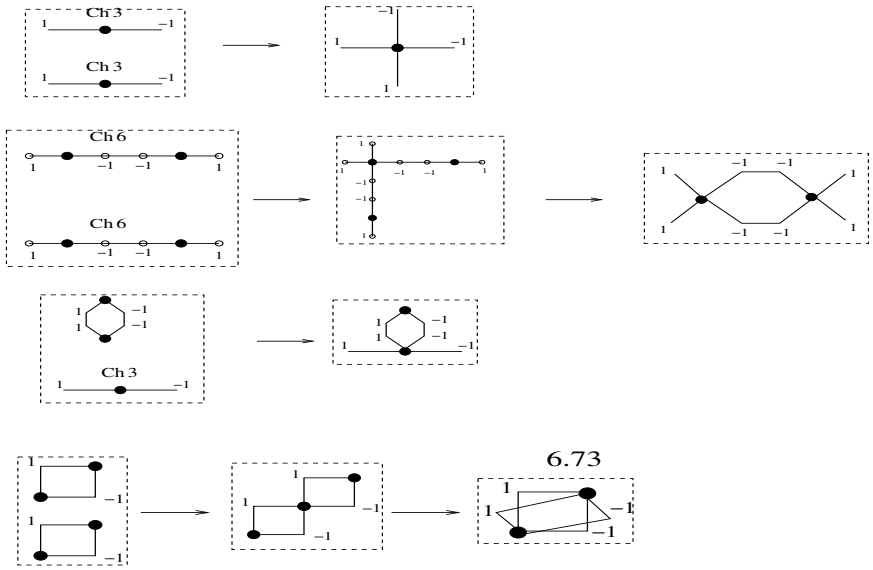


Figure 4: Examples of the soldering transform.

This transformation is valid for graphs with arbitrary weights.

4.4 Regular expansion of a graph

We have the following theorem.

Theorem 4.5 *Let x be an eigenvector of a graph G for λ and let i be a vertex connected only to p soft nodes. Let G' be the graph obtained from G by replacing i by a d-regular graph whose k vertices are all connected to the p soft nodes. Then $\lambda = p$ and an eigenvector x' of G' is formed by assigning to the new vertices, the value $x'_j = x_i/k$.*

Proof. Without loss of generality, we can assume that $i = n$ and that the p soft

nodes are $n-p+1,\ldots,n-1$. We have

$$\begin{pmatrix} \ldots & \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ 0 & \ldots & 0 & -1 & \ldots & -1 & p \end{pmatrix} \begin{pmatrix} \ldots \\ 0 \\ 0 \\ x_n \end{pmatrix} = \lambda \begin{pmatrix} \ldots \\ 0 \\ 0 \\ x_n \end{pmatrix}$$

The nth line reads

$$px_n = \lambda x_n$$

so that $\lambda = p$. The $n-1$th line reads

$$\alpha + (-1)x_n = px_{n-1} = 0$$

where α is the sum of the other terms.

Let us detail the eigenvector relation for the Laplacian for G'. Consider any new vertex j linked to the p soft nodes and to d new nodes. The corresponding line of the eigenvector relation for the Laplacian for G' reads

$$(d+p)x'_j + \sum_{i\sim j, i\geq n} (-1)x'_i = \lambda' x'_j.$$

This implies

$$(d+p-\lambda')x'_j = \sum_{i\sim j, i\geq n} x'_i.$$

An obvious solution is

$$\lambda' = \lambda = p, \quad x'_i = x'_n \ \ \forall i \geq n+1.$$

The value x'_n is obtained by examining line $n-1$. We have

$$\alpha + \sum_{i=n}^{n-k-1} (-1)x'_i = 0$$

so that

$$x'_n = \frac{x_n}{k}.$$

In fact, we can get all solutions by satisfying the two conditions

$$\forall j \geq n \ \ dx'_j = \sum_{i\sim j} x'_i, \quad x_n = \sum_{i\geq n} x'_i. \tag{16}$$

□

Fig. 5 shows examples of expansion from a single soft node for different values of d. Here the eigenvalue is 1. Fig. 6 shows examples of expansion from two soft nodes. The eigenvalue is 2. For $d=2$, the values at the edges at the bold edges are such that their sum is equal to 1. For $d=2$, the values at the

triangle are all equal to t, the same holds for the square with a value s. These values verify $3t + 4s = 1$.

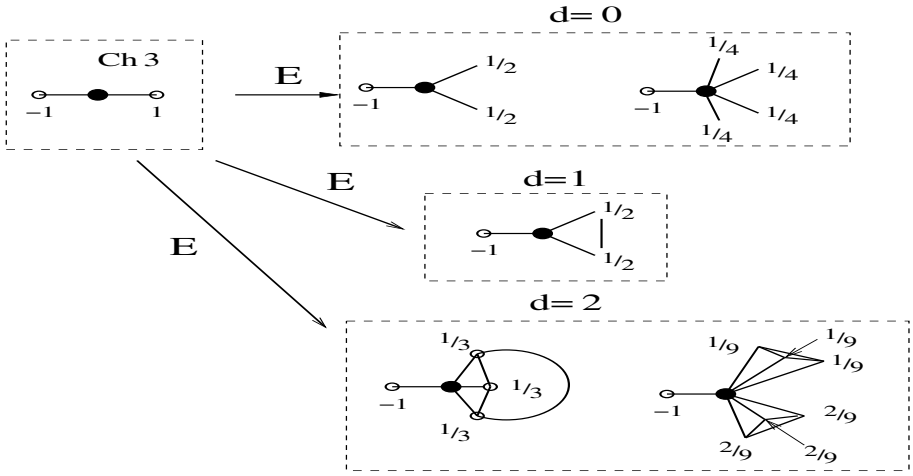


Figure 5: Examples of expansion from a single soft node.

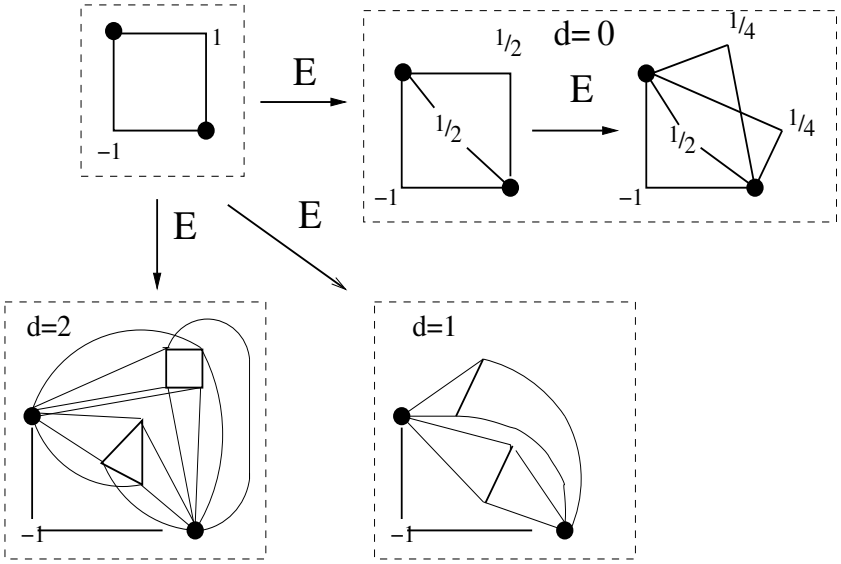


Figure 6: Examples of expansion from two soft nodes. For $d = 2$, the values at the triangle are all equal to t, the same holds for the square with a value s. These values verify $3t + 4s = 1$.

4.5 Replace coupling by square

We have the following transformation that leaves the eigenvalue unchanged [12].

Theorem 4.6 ***(Replace an edge by a soft square)***
Let x be an eigenvector of the Laplacian of a graph G for an eigenvalue λ. Let G' be the graph obtained from G by deleting a joint ij such that $x_i = -x_j$ and adding two soft vertices $k, l \in V(G')$ for the extension x' of x (i.e. $x'_m = x_m$ for $m \in V(G)$ and $x'_k = x'_l = 0$) and the four edges ik, kj, il, lj. Then, x' is an eigenvector of the Laplacian of G' for the eigenvalue λ.

This result was proved in [12] for a graph with weights 1. Here we generalize it to a graph with arbitrary weights.

Proof.

The eigenvalue relation at vertex i reads

$$(d_i - \lambda)x_i = \sum_{m\sim i, m\neq j} w_{i,m}x_m + w_{i,j}x_j$$

Since $x_i = -x_j$, this implies

$$(d_i + w_{i,j} - \lambda)x_i = \sum_{m\sim i, m\neq j} w_{i,m}x_m.$$

Introducing the two new vertices k, l such that $x'_k = x'_l = 0$ connected to i by edges of weights $w_{i,k} = \alpha w_{i,j}$, $w_{i,l} = (1-\alpha)w_{i,j}$, the relation above leads to

$$(d_i + w_{i,k} + w_{i,l} - \lambda)x'_i = \sum_{m\sim i} w_{i,m}x'_m + w_{i,k}x'_k + w_{i,l}x'_l,$$

which shows that x' is eigenvector of the new graph.

□

See Fig. 7 for an illustration of the theorem.

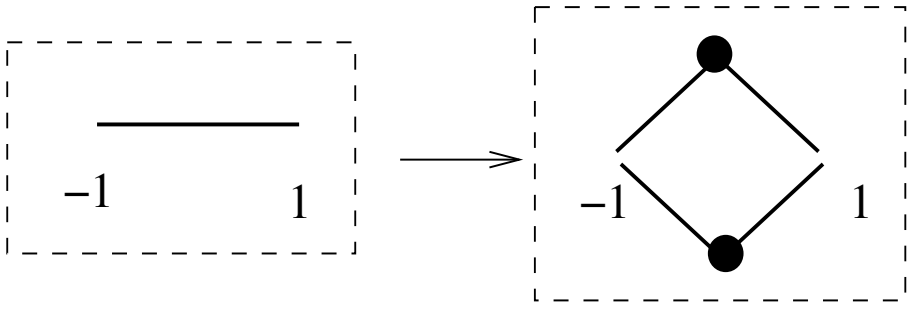


Figure 7: Replacement of coupling by a square, in both cases the eigenvalue is $\lambda = 2$.

5 Transversality : change of eigenvalue

Here we present operators that change the eigenvalue of a graph Laplacian in a predictable way. The operators shift the eigenvalue λ to $\lambda + 1$ for the first two and $\lambda + 2$ for the third one. At the end of the section we introduce the eigenvalue of a product graph.

5.1 Inserting soft nodes

Theorem 5.1 *Let x be an eigenvector of a graph G with weights 1 for λ. Assume we can pair the non zero components of x as $\{i, j\}$ where $x_i = -x_j$ non zero. Let G' be the graph obtained from G by including k soft nodes between each pair $\{i, j\}$. The vector x' so obtained is an eigenvector of the Laplacian of G' for eigenvalue $\lambda + k$.*

Proof. Let $i, j \in V(G)$ be a pair such that $x_i = -x_j$. The eigenvector equation reads

$$d_i x_i - \sum_{m \sim i} x_m = \lambda x_i.$$

Introducing k new vertices $x'_p = 0, \quad p = 1, \ldots k$ we can write the relation as

$$(d_i + k)x'_i - \sum_{m \sim i} x'_m = (\lambda + k)x'_i.$$

This shows that x' is an eigenvector for the new graph. □

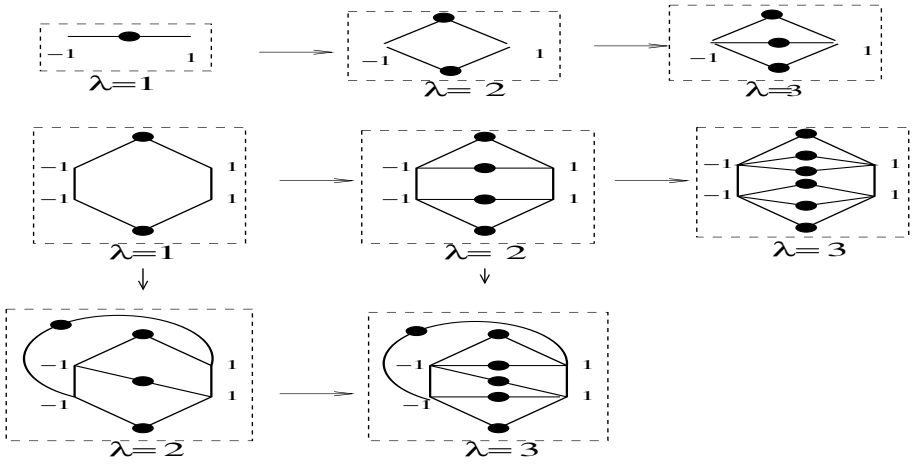


Figure 8: Example of the action of inserting a soft node.

Fig. 8 shows an example of the action of inserting a soft node.

When the graph is weighed, the result is still valid. Consider that we add only one soft vertex connected to i by a weight $w_{i,k}$. The eigenvalue of the new graph is $\lambda + w_{i,k}$.
This can transform a graph with an integer eigenvalue to a graph with an irrational eigenvalue.

5.2 Addition of a soft node

Connecting a soft node to all the vertices of a graph augments all the non zero eigenvalues by 1. This result was found by Das [11]. We recover it here and present it for completeness.

Theorem 5.2 Addition of a soft node *: Let $G(V, E)$ be a graph affording an eigenvalue $\lambda \neq 0$ for an eigenvector x. Then the new graph G' obtained by adding a node connected to all the nodes of G has eigenvalue $\lambda + 1$ for the eigenvector x' obtained by extending x by a zero component.*

See Fig. 9 for examples.

Proof. Assume λ to be an eigenvalue with eigenvector v for the Laplacian $L(G)$ of a graph G with n vertices. Now add an extra vertex $n+1$ connected to all vertices of G and form $L(G \cup \{n+1\})$. We have the following identity

$$\left(\begin{array}{c|c} & -1 \\ L(G) + I_n & -1 \\ & -1 \\ \hline -1-1, \cdots -1 & n \end{array}\right) \left(\begin{array}{c} v \\ \\ \hline 0 \end{array}\right) = (\lambda+1) \left(\begin{array}{c} v \\ \\ \hline 0 \end{array}\right)$$

which proves the statement. □

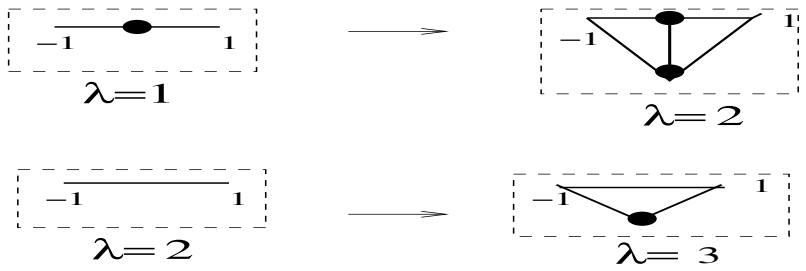


Figure 9: Examples of the addition of a soft node.

Important examples are the ones formed with the special graphs considered above. There, adding a vertex to an $n-1$ graph, one knows explicitly $n-1$ eigenvectors and eigenvalues.

The theorem 3.2 by Das [11] can be seen as a direct consequence of adding a soft node and an articulation to a graph.

5.3 Inserting a matching

First we define perfect and alternate perfect matchings.

Definition 5.3 (Perfect matching) *A perfect matching of a graph G is a matching (i.e., an independent edge set) in which every vertex of the graph is incident to exactly one edge of the matching.*

Definition 5.4 (Alternate perfect matching) *An alternate perfect matching for a vector v on the nodes of a graph G is a perfect matching for the nonzero nodes such that edges e_{ij} of the matching satisfy $v_i = -v_j \ \ (\neq 0)$.*

We have the following result [12] inspired by the alternating principle of Merris [10].

Theorem 5.5 (Add/Delete an alternate perfect matching) *Let v be an eigenvector of $L(G)$ affording an eigenvalue λ. Let G' be the graph obtained from G by adding (resp. deleting) an alternate perfect matching for v. Then, v is an eigenvector of $L(G')$ affording the eigenvalue $\lambda + 2$ (resp. $\lambda - 2$).*

This is a second operator which shifts eigenvalues by ± 2. Examples are given in Fig. 10.

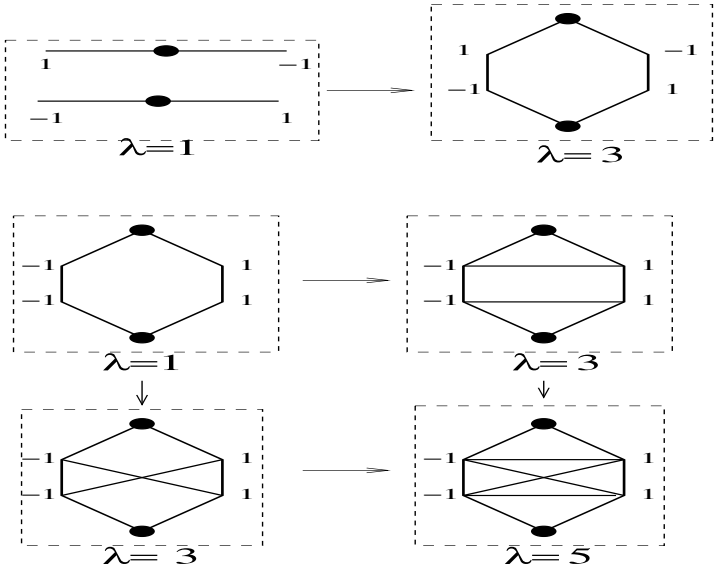


Figure 10: Examples of inserting a matching.

5.4 Cartesian product

The cartesian product $G\Box H$ of two graphs $G = (V, E)$ and $H = (W, F)$ has set of vertices $V \times W = \{(v, w), v \in V,\ w \in W\}$. It's set of edges is $\{\{(v_1, w_1), (v_2, w_2)\}\}$ such that $v_1\ v_2 \in V$ and $w_1 w_2 \in W$. We have the following result, see Merris [10].

Theorem 5.6 *If x is an eigenvector of G affording μ and y is an eigenvector of H affording ν, then the Kronecker product of x andy , $x \otimes y$ is an eigenvector of $G\Box H$ for the eigenvalue $\mu + \nu$.*

Fig. 11 illustrates the theorem.

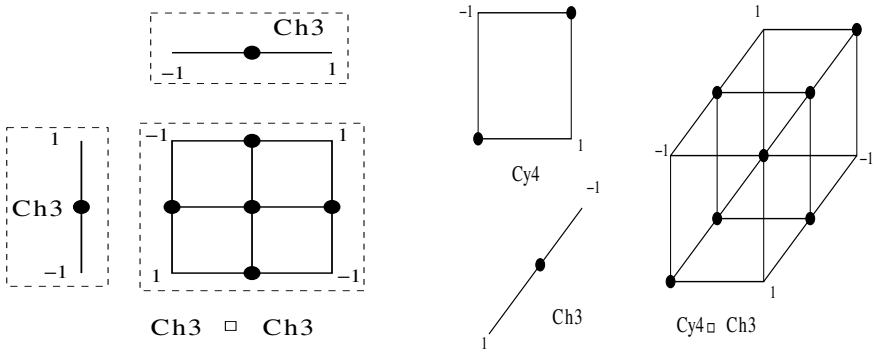


Figure 11: Cartesian product of two chains 3 (left) and of a cycle 4 and a chain 3 (right).

Important examples are the ones formed with the special graphs considered above. There, one knows explicitly the eigenvectors and eigenvalues. For example, the cartesian product $C_n \times C_m$ of two chains C_n and C_m with n and m nodes respectively has eigenvalues

$$\lambda_{i,j} = \lambda_i + \lambda_j,$$

where λ_i (resp. λ_j) is an eigenvalue for C_n (resp. C_m). The eigenvectors are

$$v^{i,j} = \cos[\frac{\pi(i-1)}{n}(p-\frac{1}{2})]\cos[\frac{\pi(j-1)}{m}(q-\frac{1}{2})],$$

where $i, p \in \{1, \ldots, n\}, \quad j, q \in \{1, \ldots, m\}$.

5.5 Graph complement

We recall the definition of the complement of a graph G.

Definition 5.7 (Complement of a graph) *Given a graph $G(V, E)$ with n vertices, its complement G^c is the graph $G^c(V, E^c)$ where E^c is the complement of E in the set of edges of the complete graph K_n.*

We have the following property, see for example [1].

Theorem 5.8 *If x is an eigenvector of a graph G with n vertices affording $\lambda \neq 0$, then x is an eigenvector of G_c affording $n - \lambda$.*

An example is shown in Fig. 12. The eigenvalues and eigenvectors are given in table 2.

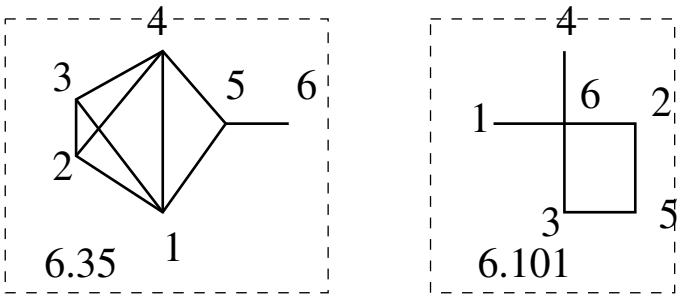


Figure 12: Graph 6.35 (left) in the classification [1] and its complement 6.101 (right).

6.35	5.2361	5.	4	3	0.7639	0
6.101	0.7639	1	2	3	5.2361	0
	0.51167	0.70711	0.	0.18257	-0.19544	0.40825
	-0.31623	0.	0.70711	-0.36515	-0.31623	0.40825
	-0.31623	0.	-0.70711	-0.36515	-0.31623	0.40825
	0.51167	-0.70711	0.	0.18257	-0.19544	0.40825
	-0.51167	0.	0.	0.73030	0.19544	0.40825
	0.12079	0.	0.	-0.36515	0.82790	0.40825

Table 2: *Eigenvalues (top lines) and eigenvectors for the two complementary graphs 6.35 and 6.101 shown in Fig. 12*

Many times, G_c is not connected. An example where G_c is connected is the cycle 6....

6 λ-soft graphs

6.1 Definitions and properties

We introduce the notions of λ, λ soft and λ soft minimal graphs. The transformations of the previous section will enable us to prove the relation between these two types of graphs.

Definition 6.1 *A graph G affording an eigenvector X for an eigenvalue λ is a λ-graph.*

Definition 6.2 *A λ-graph G affording an eigenvector X for the eigenvalue λ is λ soft if one of the entries of X is zero.*

Definition 6.3 *A graph G affording an eigenvector X for an eigenvalue λ is λ minimal if it is a λ-graph and minimal in the sense of inclusion.*

Clearly, for a given λ, there is at least one λ minimal graph. As an example the 1 soft minimal graph is shown below.

Ch 3 

6.2 λ subgraph

In the following section, we study the properties of a λ subgraph G included in a λ graph $G"(V", E")$. Consider two graphs $G(V, E)$ with n vertices and $G'(V" - V, E')$ such that E only connects elements of V and E' only connects elements of V'. Assume two graphs $G(n)$ and $G'(n')$ are included in a large graph $G"$ and are such that $G(V, E)$ Assume p vertices of G are linked to p' vertices of G'. We label the p vertices of G, $n-p+1, \ldots, n$ and the p' vertices of G', $1, \ldots, p'$. We have

$$LX = \lambda X, \tag{17}$$

$$L"X" = \lambda X", \tag{18}$$

where $L"$ is the graph Laplacian for the large graph $G"$; $L"$ can be written as

$$L" = \begin{pmatrix} L & 0 \\ 0 & L' \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & a & -b & 0 \\ 0 & -b^T & c & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

A first result is

Theorem 6.4 *The square matrix $\delta = \begin{pmatrix} a & -b \\ -b^T & c \end{pmatrix}$ is a graph Laplacian.*

Proof. The submatrices a, b, c have respective sizes $a(p,p)$, $b(p,p')$, $c(p',p')$, a and c are diagonal and verify

$$a_{ii} = \sum_{j=1}^{p'} b_{ij}, \quad c_{ii} = \sum_{j=1}^{p} b_{ji}. \tag{19}$$

In other words

$$a\hat{1}_p = b\hat{1}_{p'}, \quad c\hat{1}_{p'} = b^T\hat{1}_p,$$

where $\hat{1}_p$ is a p column vector of 1. □

At this point, we did not assume any relation between the eigenvectors X for G and X" for G". We have the following

Theorem 6.5 *The eigenvalue relations (17,18) imply either $X = X"(1:n)$ or $X(1:n-p) = 0$.*

Proof. For $p = 1$ and λ a single eigenvalue, $\text{rank}(L - \lambda I) = n - 1$ so either $X = X"(1:n)$ or $X(1:n-1) = 0$.
We admit the result for $p > 1$. □

We can then assume that the eigenvectors of L" have the form

$$L" \begin{pmatrix} X \\ X' \end{pmatrix} = \lambda \begin{pmatrix} X \\ X' \end{pmatrix},$$

where $LX = \lambda X$. Substituting L", we get

$$\lambda \begin{pmatrix} X \\ X' \end{pmatrix} = \begin{pmatrix} L & 0 \\ 0 & L' \end{pmatrix} \begin{pmatrix} X \\ X' \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & a & -b & 0 \\ 0 & -b^T & c & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} X \\ X' \end{pmatrix}.$$

Using the relation (17) we obtain

$$\begin{pmatrix} 0 & 0 \\ 0 & a \end{pmatrix} X + \begin{pmatrix} 0 & 0 \\ -b & 0 \end{pmatrix} X' = 0, \tag{20}$$

$$L'X' - \begin{pmatrix} 0 & b^T \\ 0 & 0 \end{pmatrix} X + \begin{pmatrix} c & 0 \\ 0 & 0 \end{pmatrix} X' = \lambda X'. \tag{21}$$

There are p non trivial equations in the first matrix equation and p' in the second one. Using an array notation (like in Fortran), the system above can be written as

$$aX(n-p+1:n) - bX'(1:p') = 0, \tag{22}$$

$$-b^T X(n-p+1:n) + cX'(1:p') + (L'X')(1:p') = \lambda X'(1:p'), \tag{23}$$

$$(L'X')(p'+1:n') = \lambda X'(p'+1:n'), \tag{24}$$

Extracting X from the first equation, we obtain

$$X(n-p+1:n) = a^{-1}bX'(1:p'), \tag{25}$$

and substituting in the second equation yields the closed system in X'

$$(-b^T a^{-1} b + c)X'(1:p') + (L'X')(1:p') = \lambda X'(1:p'), \tag{26}$$

$$(L'X')(p'+1:n') = \lambda X'(p'+1:n'), \tag{27}$$

where we used the fact that the matrix a of the degrees of the connections is invertible by construction.

Theorem 6.6 *The matrix*

$$\Delta \equiv -b^T a^{-1} b + c,$$

is a generalized graph Laplacian: it is a Laplacian of a weighted graph. Its entries are rationals and not necessarily integers.

Proof. To prove this, note first that Δ is obviously symmetric. We have

$$\Delta \hat{1}_{p'} = -b^T a^{-1} b \hat{1}_{p'} + c\hat{1}_{p'} = -b^T a^{-1} a \hat{1}_p + b^T \hat{1}_p = 0.$$

This shows that the each diagonal element of Δ is equal to the sum of it's corresponding row so that Δ is a graph Laplacian. □ From theorem (2.10), the eigenvalues of Δ are integers or irrationals and correspond to eigenvectors with integer or irrational components.

We then write equations (26,27) as

$$(\bar{\Delta} + L')X' = \lambda X', \tag{28}$$

where

$$\bar{\Delta} = \begin{pmatrix} \Delta & 0 \\ 0 & 0 \end{pmatrix}$$

This is an eigenvalue relation for the graph Laplacian $(\bar{\Delta} + L')$. Four cases occur.

(i) $\lambda = 0$ then X' is a vector of equal components and X also.

(ii) $\lambda \neq 0$ is an eigenvalue of L'. Then one has the following

Theorem 6.7 *Assume a graph G" is λ for an eigenvector $X" = (X, X')^T$ and contains a λ graph G for the eigenvector X. Consider the graph G' with vertices $V(G") - V(G)$ and the corresponding edges in G".*
If G' is λ then G" is obtained from G using the articulation or link transformations.

Proof. Since $\lambda \neq 0$ is an eigenvalue of L', we can choose X' an eigenvector for λ so that $L'X' = \lambda X'$, then $\Delta X' = 0$.
A first possibility is $X' = 0$, this corresponds to an articulation between G and G'.
If $X' \neq 0$, $L'X' = \lambda X'$, implies that X' is not a vector of equal components so that $X' \notin \text{Null}(\Delta)$. The only possibility for $\Delta X' = 0$ is $\Delta = 0$ so that

$$c = b^T a^{-1} b.$$

The term $(b^T a^{-1} b)_{ij}$ is

$$(b^T a^{-1} b)_{ij} = \sum_{k=1}^{p} \frac{b_{ki} b_{kj}}{a_{kk}}.$$

Since the matrix c is diagonal, we have

$$\sum_{k=1}^{p} \frac{b_{ki} b_{kj}}{a_{kk}} = 0, \forall i \neq j$$

Then $b_{ki} b_{kj} = 0$ so that a vertex k from G is only connected to one other vertex i or j from G'. Then $p = p'$. This implies $a_{ii} = c_{ii} = 1, \forall i \in \{1, , \ldots, p\}$. The graphs G and G' are then connected by a number of edges between vertices of same value. □

(iii) $\lambda \neq 0$ is not an eigenvalue of L' and L' and $\bar{\Delta}$ share a common eigenvector X' for eigenvalues λ' and $\lambda - \lambda' > 0$.
For $\lambda - \lambda' = 1$, a possibility is to connect a soft node of G to G'. For $\lambda - \lambda' = p$ integer, a possibility is to connect p soft nodes of G to G'.
We conjecture that there are no other possibilities.

(iv) $\lambda \neq 0$ is not an eigenvalue of L' and L' and $\bar{\Delta}$ have different eigenvectors. Then there is no solution to the eigenvalue problem (28).
To see this, assume the eigenvalues and eigenvectors of L' and $\bar{\Delta}$ are respectively ν_i, V^i, μ_i, W^i so that

$$L' V^i = \nu_i V^i, \quad \bar{\Delta} W^i = \mu_i W^i, \quad i = 1, 2, \ldots n$$

The eigenvectors can be chosen orthonormal and we have

$$QV = WQ$$

where $Q = (q_k^j)$ is an orthogonal matrix, V and W are the matrices whose columns are respectively V^i and W^i. We write

$$W^j = \sum_k q_k^j V^k.$$

Assuming X' exists, we can expand it as $X' = \sum_i \alpha_i V^i$ Plugging this expansion intro the relation $(\bar{\Delta} + L')X' = \lambda X'$ yields

$$\sum_i \left(\alpha_i \nu_i V^i + \alpha_i \sum_j q_j^i \mu_j \sum_k q_k^j V^k \right) = \sum_i \lambda \alpha_i \nu_i V^i$$

Projecting on a vector V^m we get

$$\alpha_m \nu_m + \alpha_m \sum_j q_j^m \mu_j q_m^j = \lambda \alpha_m \nu_m$$

A first solution is $\alpha_m = 0, \forall m$ so that $X' = 0$, an articulation. If $\alpha_m \neq 0$ then we get the set of linear equations linking the ν_i to the μ_i.

$$\sum_j q_j^m \mu_j q_m^j = (\lambda - 1)\nu_m, \quad m = 1, \ldots n$$

Since Q is a general orthogonal matrix, the terms q_j^m are irrational in general. Therefore we conjecture that there are no solutions.

6.3 Examples of λ subgraphs

Using simple examples, we illustrate the different scenarios considered above. We first consider theorem (6.7), see Fig. 13.

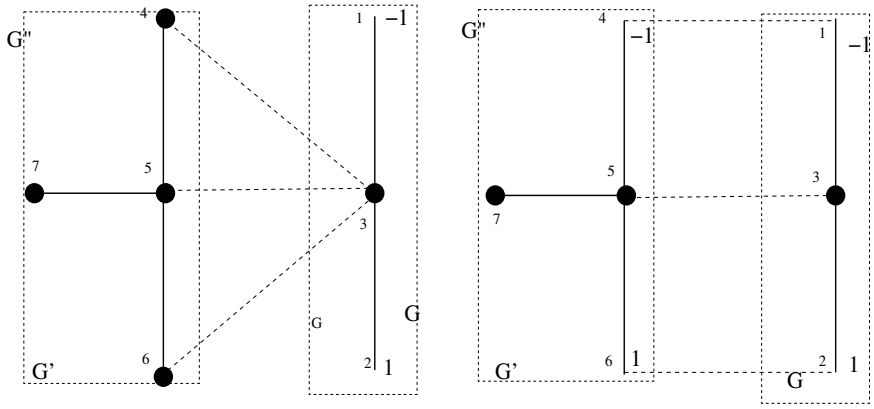


Figure 13: Two configurations where a graph G is included in a larger graph G" for the eigenvalue 1.

Consider the configuration on the left of Fig. 13. We have

$$L = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ -1 & -1 & 2 \end{pmatrix}, \quad L' = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 3 & -1 & -1 \\ 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}. \tag{29}$$

Note that L and L' have 1 as eigenvalue. Here $p = 1, p' = 3$ and

$$a = 3, b = (1,1,1)^T, c = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

so that

$$\Delta = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \end{pmatrix}.$$

The matrices $\bar{\Delta}$ and L' have different eigenvectors for the same eigenvalue 1. Choosing X' an eigenvector of L' for the eigenvalue 1 yields $\bar{\Delta} X' = 0$. The only solution is $X' = 0$, this is an articulation.

For the configuration on the right of Fig. 13 we have $p = p' = 3$.

$$a = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad c = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

so that $\Delta = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. We have

$$LX = 1X, \tag{30}$$
$$L"(X, X')^T = 1(X, X')^T, \tag{31}$$

where $X = (X_1, X_2, X_3)^T$ In this configuration, X' is an eigenvector of L' for the eigenvalue 1. and we have Link connections between G and G'.

Finally, we show an example of case (iii) where $G, G"$ are 2 soft and G' is 1 soft.

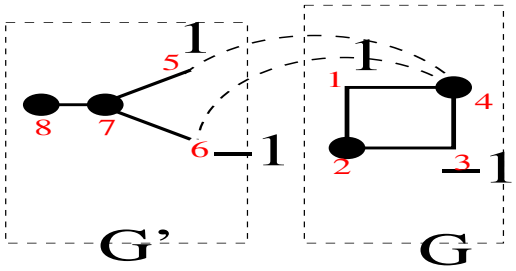


Figure 14: An example of case (iii) for eigenvalue $\lambda = 2$.

We have to solve $(\bar{\Delta} + L')X' = 2X'$ where

$$L = \begin{pmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}, \quad L' = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}, \quad \bar{\Delta} = \begin{pmatrix} 0.5 & -0.5 & 0 \\ -0.5 & 0.5 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Note that the eigenvector $X' = (1, -1, 0, 0)^T$ is shared by L' and $\bar{\Delta}$ so that $(\bar{\Delta} + L')X' = 2X'$.

The transformations introduced in the two previous sections enable us to link the different members of a given class. To summarize, we have

- Articulation : one can connect any graph G_2 to the soft nodes of a given graph G_1 and keep the eigenvalue. The new graph $G_1 \cup G_2$ has soft nodes everywhere in G_2.

- Link : introducing a link between equal nodes does not change the eigenvalue and eigenvector.
- Contraction of a d-regular graph linked to a soft node. To have minimal graphs in the sense of Link we need to take $d = 0$.
- Soldering : one can connect two graphs by contracting one or several soft nodes of each graph.

In the next subsections we present a classification of small size λ soft graphs for different λs.

6.4 1-soft graphs

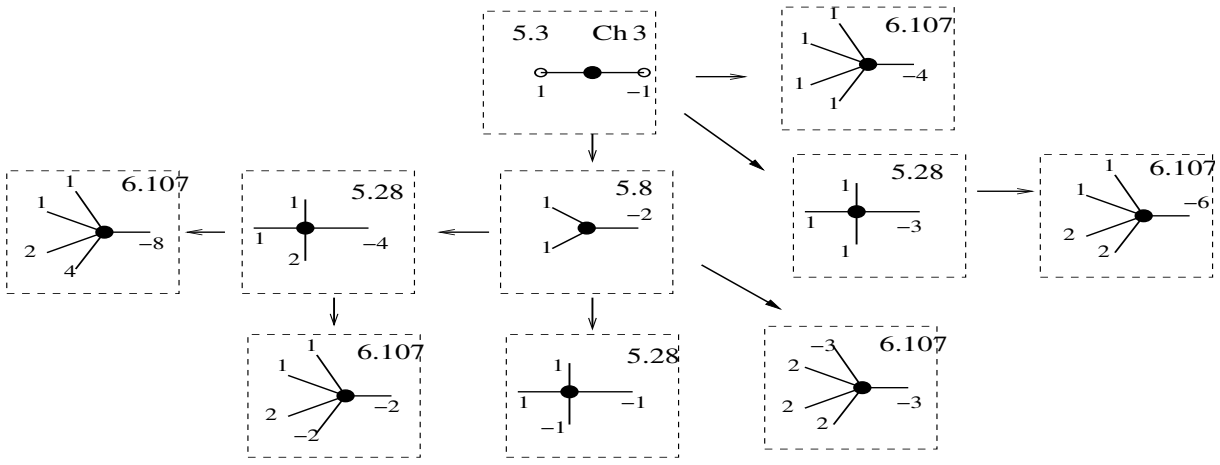


Figure 15: 1_s graphs: graphs generated by expansion.

Fig. 15 shows some of the 1s graphs generated by expansion. Note the variety of possibilities.

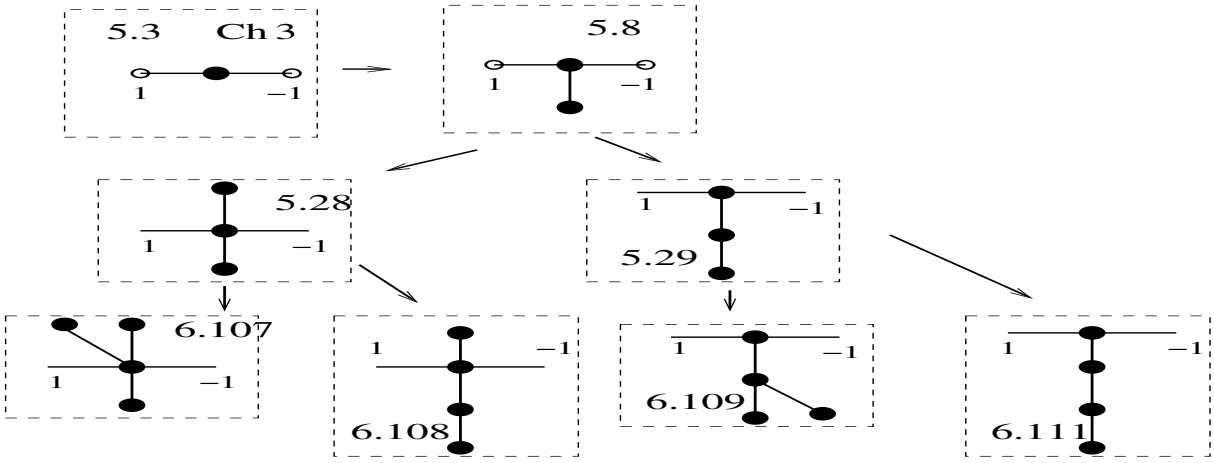


Figure 16: 1_s graphs: graphs generated by articulation

Fig. 16 shows some of the 1s graphs generated by articulation. The $1, 0, -1$ configuration remains clearly visible.

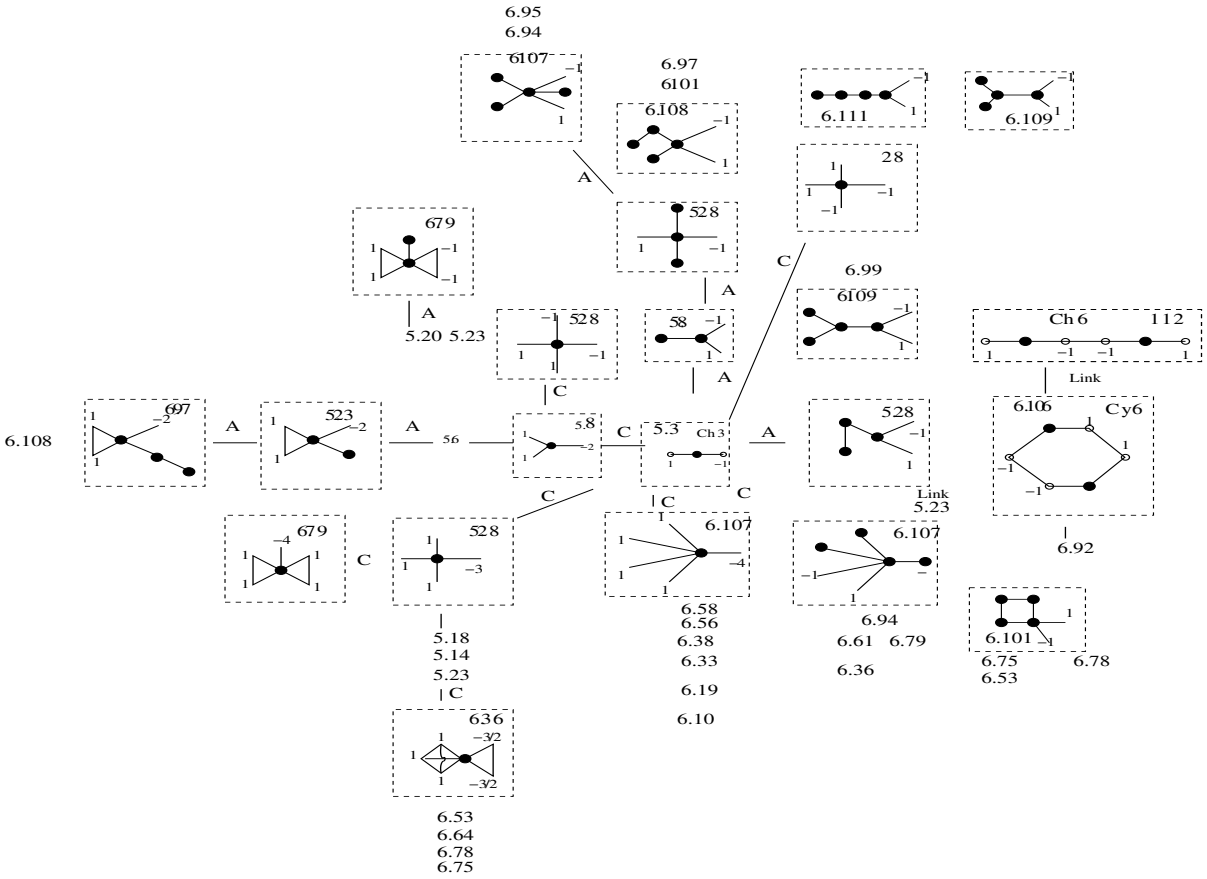


Figure 17: 1-soft graphs. The soft nodes are in boldface. We only present symmetric expansions so that links are possible.

Fig. 17 shows the 1s graphs with at most 6 vertices. Notice how they are linked by articulation (A), expansion/contraction (C) and links and can all be obtained from the graph 5.3 (chain 3). The connection $Ch3$ - 28 is a contraction of two $Ch3$ chains. Connecting two 3 chains $Ch3$ with an Link transformation we obtain a chain 6 $Ch6$. One can also go from $Ch6$ to 23 by soldering the two soft nodes.

6.5 2-soft graphs

Fig. 18 shows some of the 2s graphs generated by expansion of the 5.7 graph.

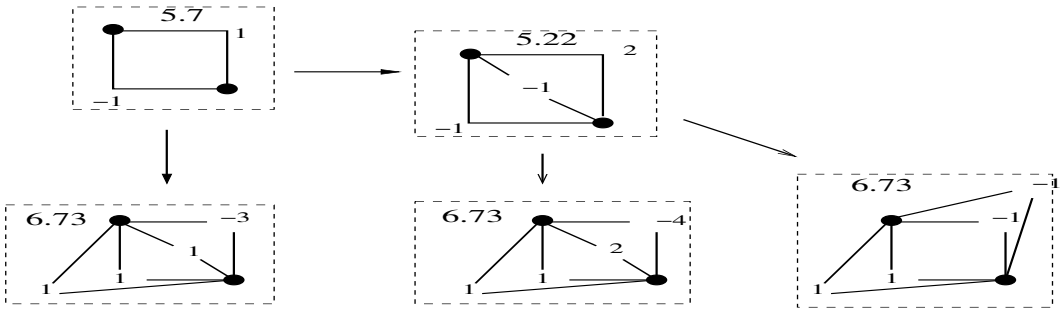


Figure 18: 2_s graphs: graphs generated by expansion.

Similarly Fig. 19 shows some of the 2s graphs generated by articulation from the same graph.

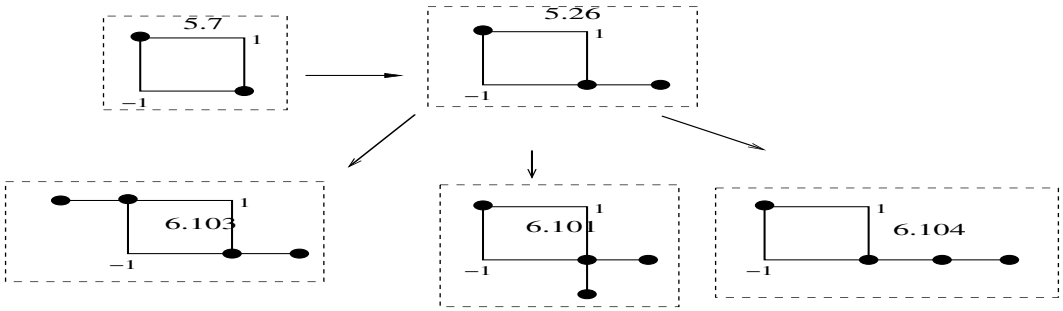


Figure 19: 2_s graphs: graphs generated by articulation

Fig. 20 shows all 2s graphs with at most 6 vertices. We included graph 5.1 because with a link it gives configuration 6.104. Notice how all graphs can be generated from 5.5 and 5.1.

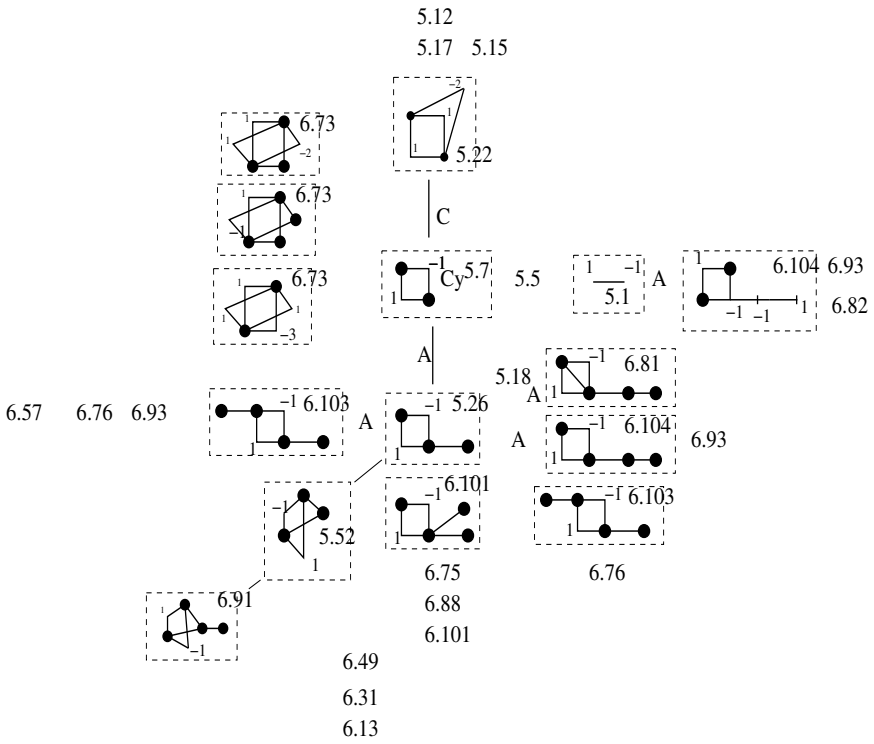


Figure 20: 2-soft graphs

6.6 3-soft graphs

Fig. 21 shows a 3s graph generated by expansion of graph 5.22.

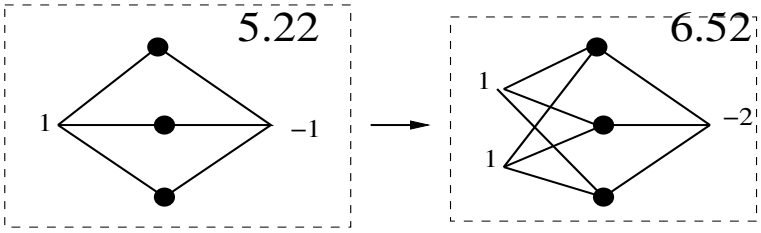


Figure 21: 3_s graphs: graphs generated by expansion.

Fig. 22 shows some 3s graphs generated by articulation on graphs 5.2 and 5.22.

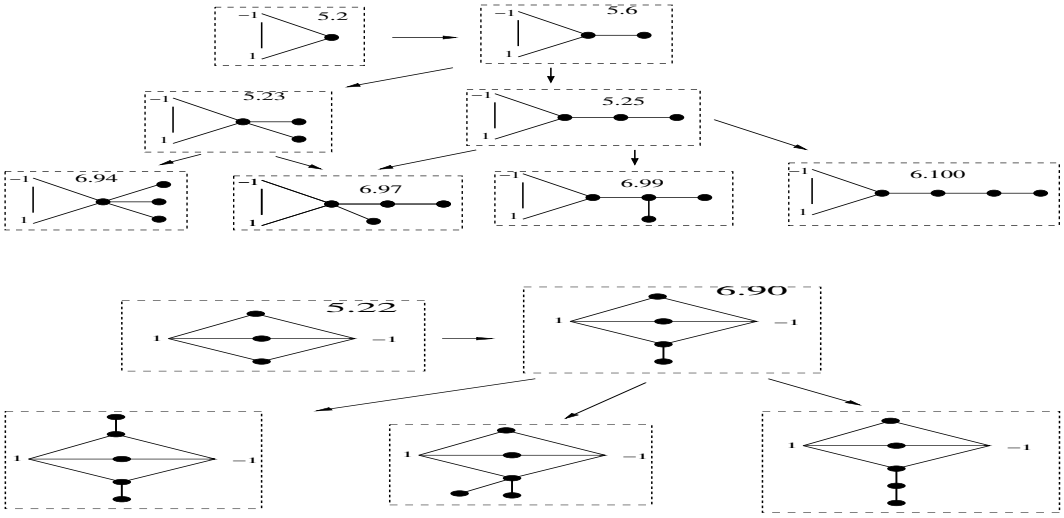


Figure 22: 3_s graphs: graphs generated by articulation

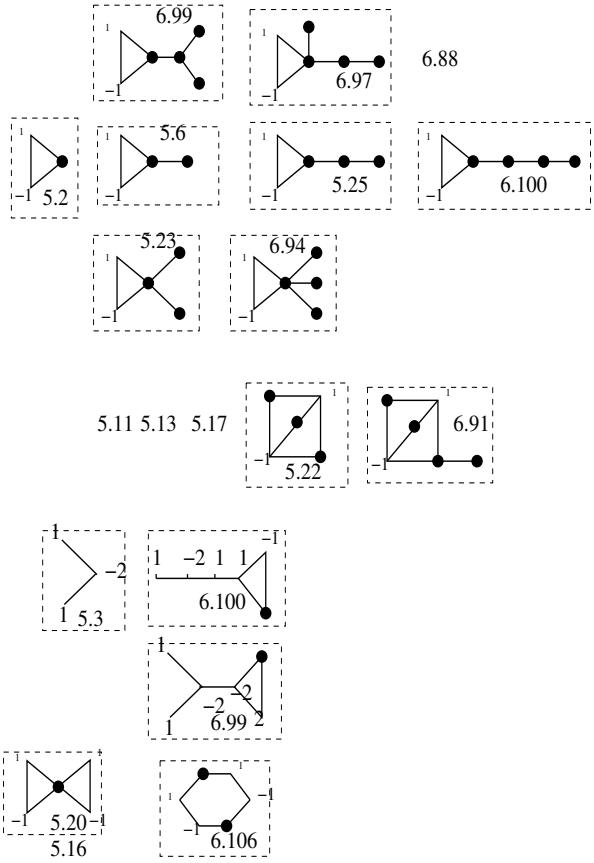


Figure 23: 3-soft graphs.

Fig. 23 shows all 3s graphs with at most 6 vertices. Notice how they are generated by graphs 5.2, 5.22 and 5.3. Graph 5.20 is the soldering of two graphs 5.2 .

6.7 4-soft graphs

Fig. 24 shows some 4s graphs generated by articulation on the graph 5.3.

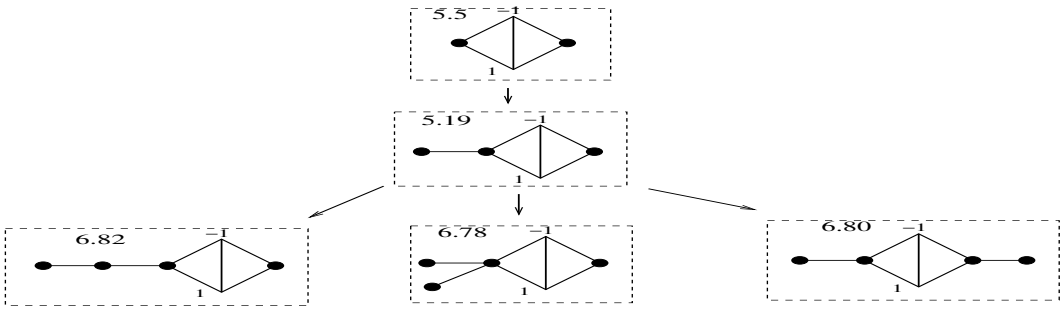


Figure 24: 4_s graphs: graphs generated by articulation

Fig. 25 shows the 4s graphs with at most 6 vertices. Notice how they are generated from graphs 5.5 (2 configurations) and 6.93. The graph 5.7 is included to show its connection to 6.93 (replacing a matching by a square).

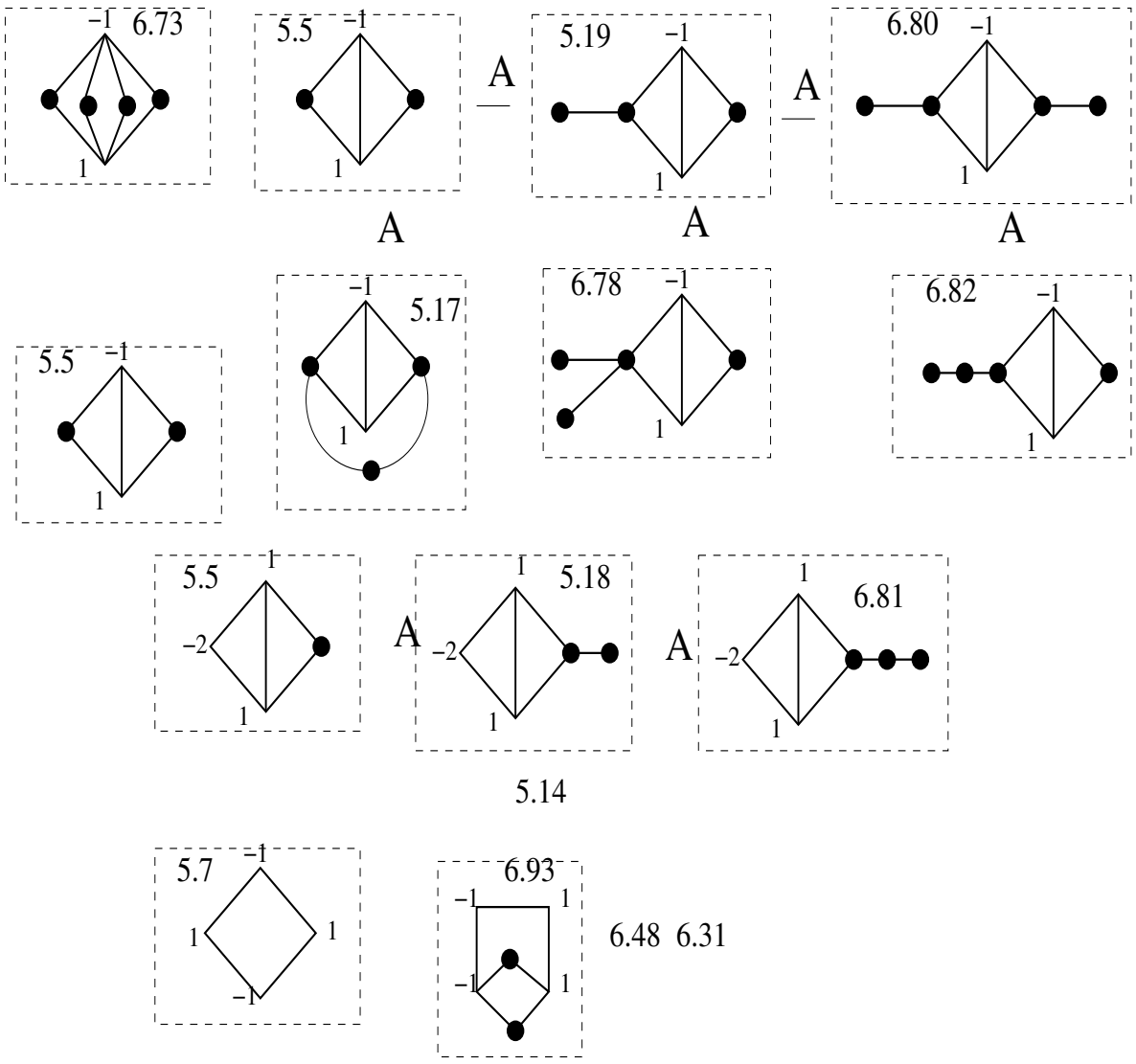


Figure 25: 4-soft graphs.

6.8 5-soft graphs

Fig. 26 shows 5s graphs with at most 6 vertices. Notice how they stem from graphs 6.70, 5.13 and two configurations of 5.15.

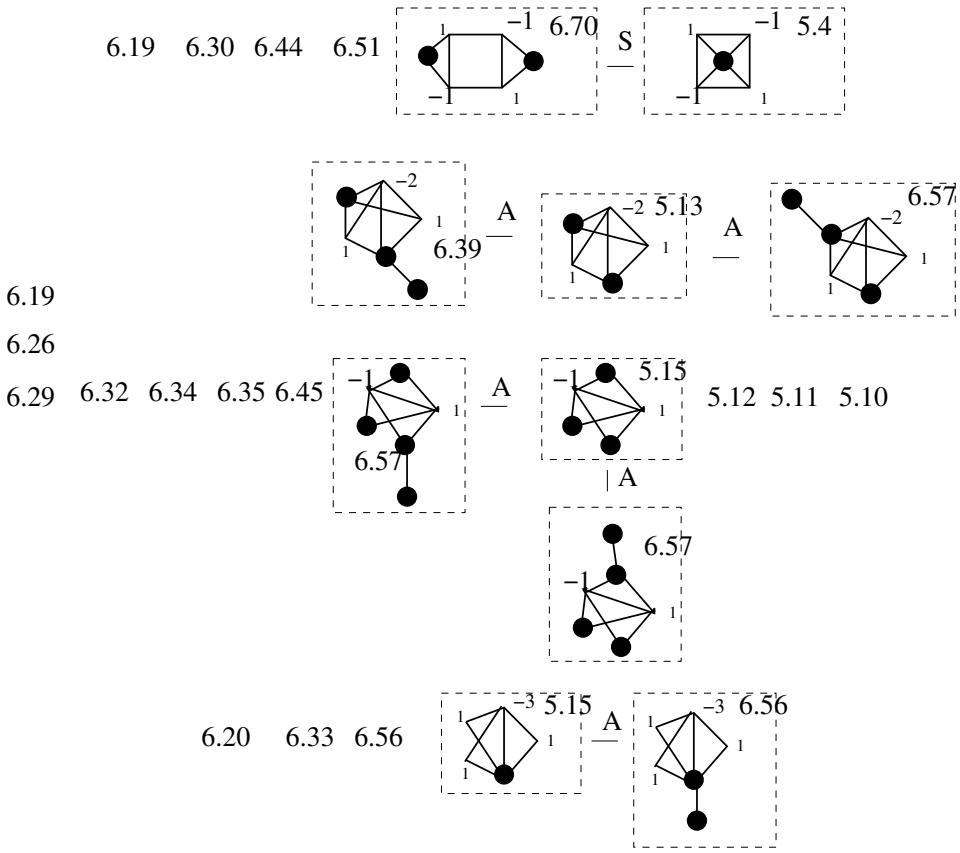


Figure 26: 5-soft graphs.

6.9 6-soft graphs

Fig. 27 shows 6s graphs with at most 6 vertices. Notice how these graphs stem from graphs 6.9, 6.37, 6.2 (two configurations) and 6.16.

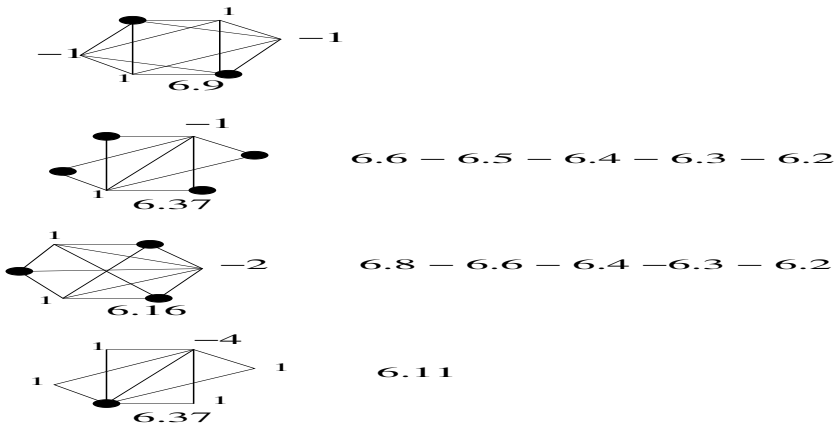


Figure 27: 6-soft graphs.

6.10 x-soft graphs, x non integer

As proven above, the only eigenvalues that are non integer are irrational. For these, there can be soft nodes. Among the 5 node graphs, we found irrational eigenvalues for the chain 5 and the cycle 5. In addition, there are the following

nb. in classification	eigenvalue	eigenvector
5.16	$\lambda_2 = 3 - \sqrt{2}$	$(-0.27, -0.65, 0, 0.65, 0.27)^T$
5.16	$\lambda_4 = 3 + \sqrt{2}$	$(0.65, -0.27, 0, 0.27, -0.65)^T$
5.21	$\lambda_4 = (7 + \sqrt{5})/2$	$(-0.6, 0.6, 0.37, 0, -0.37)^T$
5.21	$\lambda_5 = (7 - \sqrt{5})/2$	$(-0.37, 0.37, -0.6, 0, 0.6)^T$
5.24	$\lambda_2 = (5 - \sqrt{13})/2$	$(-0.67, -0.2, 0.2, 0.67, 0)^T$
5.24	$\lambda_5 = (5 + \sqrt{13})/2$	$(-0.2, 0.67, -0.67, 0.2, 0)^T$
5.30 (chain 5)	$\lambda_4 = (3 + \sqrt{5})/2$	$(-0.6, 0.6, 0.37, 0, -0.37)^T$
5.30 (chain 5)	$\lambda_5 = (3 - \sqrt{5})/2$	$(-0.37, 0.37, -0.6, 0, 0.6)^T$

Table 3: *Non trivial graphs with soft nodes and non integer eigenvalues.*

Remarks
The graph 5.16 is 3 soft. The graphs 5.21 and 5.24 are not part of an integer soft class. They are

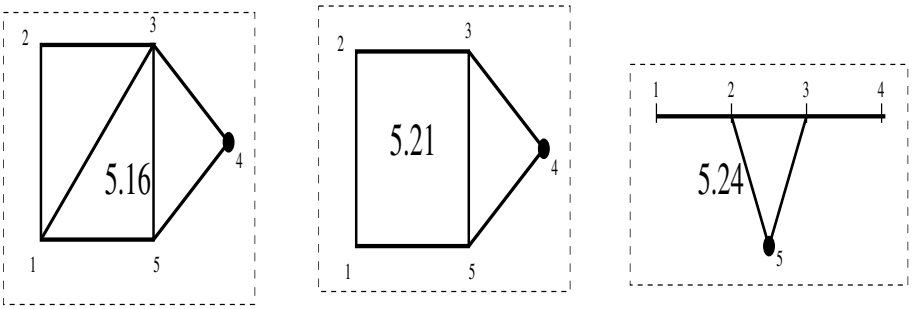


Figure 28: The graphs 5.16, 5.21 and 5.24 with their soft node

- Graph 5.16 is a chain 4 with a soft node added.
- Graph 5.21 is obtained from chain 5 (graph 5.30) by inserting a soft node.

6.11 Minimal λ soft graphs

We computed the minimal λ soft graphs for $\lambda = 1, \ldots, 6$. These are presented in Fig. 29.

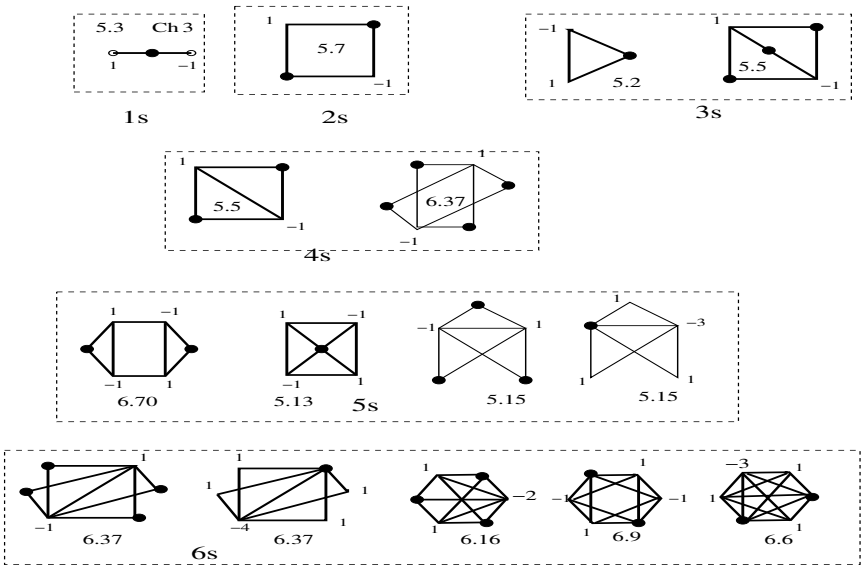


Figure 29: The minimal λ soft graphs for $\lambda = 1, 2, 3, 4, 5$ and 6.

Note that there is a unique minimal λ-soft graph for $\lambda = 1$ and 2. There

are two minimal 3-soft graphs and 4-soft graphs. There are four minimal 5-soft graphs. The first two are generated by respectively inserting a soft node and adding a soft node to the minimal 4-soft graph. The third and fourth ones are obtained respectively by adding three soft nodes to the 2 clique and adding a soft node to the 4 star.

Three systematic ways to generate minimal $\lambda+1$-soft graphs are (i) inserting a zero to a λ-soft graph,
(ii) adding a zero to aλ-soft graph and
(iii) adding a matching to a $\lambda - 1$-soft graph. One can therefore generate systematically minimal 7-soft, 8-soft.. graphs.

7 Conclusion

We reviewed families of graphs whose spectrum is known and presented transformations that preserve an eigenvalue. The link, articulation and soldering were contained in Merris [10] and we found two new transformations : the regular expansion and the replacement of a coupling by a square. We also showed transformations that shift an eigenvalue : insertion of a soft node (+1), addition of a soft node (+1), insertion of a matching (+2). The first is new and the second and third were found by Das [11] and Merris [10] respectively.

From this appears a landscape of graphs formed by families of λ-graphs connected by these transformations. These structures remain to be understood. We presented the connections between small graphs with up to six vertices. Is it possible to obtain all the λ graphs using a series of elementary transformations? Or just part of these ?

We answered partially the question: can one predict eigenvalues/eigenvectors from the geometry of a graph ? by examining the situation of a a λ subgraph G of a λ graph G". We showed that if the remainder graph G' is λ, it is an articulation or a link of G. If not and if G and G' share an eigenvector, the two may be related by adding one or several soft nodes to G'.

A number of the graphs we studied have irrational eigenvalues and we can define λ graphs for these as well because the transformations apply. However we did not find any connection between λ graphs and μ graphs if λ is an integer and μ an irrational.

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8 Appendix A: Graph classification

The following tables indicate the graph classification we used. Each line in the "connections" column is the connection list of the corresponding graph.

classification [1]	nodes	links	connections
1	2	1	12
2	3	3	12 13 23
3	3	2	12 23
4	4	6	12 13 14 23 24 34
5	4	5	12 13 14 23 34
6	4	4	12 13 23 34
7	4	4	12 14 23 34
8	4	3	12 23 24
9	4	3	12 23 34
10	5	10	12 13 14 15 23 24 25 34 35 45
11	5	9	12 13 14 15 23 24 34 35 45
12	5	8	12 14 15 23 42 25 34 45
13	5	8	12 13 15 23 24 34 35 45
14	5	7	12 13 14 23 24 34 35
15	5	7	13 15 23 25 34 35 45
16	5	7	12 13 15 23 34 35 45
17	5	7	12 14 15 23 25 34 45
18	5	6	12 13 14 23 34 35
19	5	6	12 14 23 24 34 35
20	5	6	12 13 23 34 35 45
21	5	6	12 15 23 34 35 45
22	5	6	13 15 23 25 34 45
23	5	5	12 13 23 34 35
24	5	5	12 23 25 35 34
25	5	5	12 13 23 34 45
26	5	5	12 14 23 34 35
27	5	5	12 15 23 34 45
28	5	4	13 23 34 35
29	5	4	12 13 14 45
30	5	4	12 23 34 45

Table 4: *Graphs of less than 5 nodes labelled* 1 *to* 30 *in classification [1].*

classification [1]	nodes	links	connections
1	6	15	12 13 14 15 16 23 24 25 26 34 35 36 45 46 56
2	6	14	12 13 15 16 23 24 25 26 34 35 36 45 46 56
3	6	13	12 14 15 16 23 24 25 26 34 35 45 46 56
4	6	13	12 13 15 16 23 24 25 26 34 35 45 46 56
5	6	12	12 13 15 16 23 25 26 34 35 36 45 56
6	6	12	12 13 14 15 16 23 25 34 35 36 45 56
7	6	12	12 13 15 16 23 24 25 26 34 35 45 56
8	6	12	12 13 15 16 23 24 34 35 36 45 46 56
9	6	12	12 13 15 16 23 24 26 34 35 45 46 56
10	6	11	12 13 14 15 23 24 25 34 35 45 56
11	6	11	12 14 16 23 24 26 34 36 45 46 56
12	6	11	12 13 15 16 23 25 26 34 35 45 56
13	6	11	12 15 16 23 24 25 26 34 35 45 56
14	6	11	12 13 15 16 23 25 26 34 36 45 56
15	6	11	12 14 15 16 23 24 34 35 45 46 56
16	6	11	12 14 15 16 23 25 34 35 36 45 56
17	6	11	12 14 15 16 23 26 34 35 45 46 56
18	6	11	12 15 16 23 24 26 34 35 45 46 56
19	6	10	12 13 15 23 24 25 34 35 45 56
20	6	10	12 13 14 15 23 24 34 35 45 56
21	6	10	12 15 16 23 24 25 26 35 45 56
22	6	10	12 13 15 16 23 34 35 36 45 56
23	6	10	12 16 23 25 26 34 35 36 45 56
24	6	10	12 15 16 23 25 26 34 35 45 56
25	6	10	12 13 14 15 16 23 34 36 45 56
26	6	10	12 14 16 23 34 35 36 45 46 56
27	6	10	12 15 16 23 26 34 35 36 45 56
28	6	10	12 14 16 23 24 34 35 45 46 56
29	6	10	12 14 16 23 24 26 34 36 45 56
30	6	10	12 15 61 23 24 25 34 36 45 56

Table 5: *6 node graphs labelled* 1 *to* 30 *in classification [1].*

classification [1]	nodes	links	connections
31	6	10	12 15 16 23 24 26 34 35 45 56
32	6	10	12 14 16 23 25 26 34 36 45 56
33	6	9	12 15 23 24 25 34 35 45 56
34	6	9	12 14 15 23 24 25 34 45 56
35	6	9	12 13 14 15 23 24 34 45 56
36	6	9	12 13 14 23 24 34 45 46 56
37	6	9	12 13 14 15 16 24 34 45 46
38	6	9	12 14 15 23 25 34 35 45 56
39	6	9	12 13 15 23 24 25 34 45 56
40	6	9	12 13 16 23 34 35 36 46 56
41	6	9	12 16 23 34 35 36 45 46 56
42	6	9	12 16 23 24 26 34 45 46 56
43	6	9	12 13 16 23 34 35 36 45 56
44	6	9	12 13 16 23 34 36 45 46 56
45	6	9	12 16 23 25 34 35 36 45 56
46	6	9	12 13 15 16 23 34 36 45 56
47	6	9	12 13 16 23 25 26 34 45 56
48	6	9	12 15 16 23 26 34 35 45 56
49	6	9	12 15 16 23 24 26 35 45 56
50	6	9	12 14 15 16 23 34 36 45 56
51	6	9	12 15 24 16 23 34 36 45 56
52	6	9	12 14 16 23 25 34 36 45 56
53	6	8	12 13 14 23 24 34 45 46
54	6	8	12 13 14 23 24 25 34 36
55	6	8	12 13 14 23 24 34 45 56
56	6	8	13 15 23 25 34 35 45 56
57	6	8	12 14 23 24 25 34 45 56
58	6	8	12 15 23 25 34 35 45 56
59	6	8	12 13 15 23 34 35 45 56
60	6	8	12 14 15 23 24 34 45 56

Table 6: *6 node graphs labelled* 31 *to* 60 *in classification [1].*

classification [1]	nodes	links	connections
61	6	8	12 13 14 15 16 23 45 46
62	6	8	12 14 23 42 34 35 36 56
63	6	8	12 14 15 23 25 34 45 56
64	6	8	12 13 15 23 25 34 45 56
65	6	8	12 13 15 23 24 34 45 56
66	6	8	12 16 24 34 36 45 56 46
67	6	8	12 13 16 23 34 36 45 56
68	6	8	12 13 16 23 34 35 45 56
69	6	8	12 15 16 23 26 34 45 56
70	6	8	12 13 16 23 34 45 46 56
71	6	8	12 13 16 23 34 35 46 56
72	6	8	12 15 16 23 34 36 45 56
73	6	8	12 15 23 24 26 35 45 56
74	6	8	12 14 16 23 25 34 45 56
75	6	7	12 13 14 23 34 35 36
76	6	7	12 23 24 25 34 45 46
77	6	7	12 14 23 24 25 34 36
78	6	7	12 14 23 24 34 35 36
79	6	7	12 13 23 34 36 35 45
80	6	7	12 23 25 34 35 45 46
81	6	7	12 13 14 23 34 35 56
82	6	7	12 14 23 24 34 35 56
83	6	7	12 13 23 34 35 45 46
84	6	7	12 13 23 34 45 46 56
85	6	7	12 15 23 24 34 45 46
86	6	7	12 13 15 23 34 45 46
87	6	7	12 13 15 23 34 45 46
87B	6	7	12 13 15 24 34 45 56
88	6	7	12 14 23 34 35 36 56
89	6	7	12 15 16 23 34 45 56
90	6	7	13 15 23 25 34 45 56

Table 7: *6 node graphs labelled* 61 *to* 90 *in classification [1]. Note that 87B is absent from [1].*

classification [1]	nodes	links	connections
91	6	7	12 14 23 25 34 45 56
92	6	7	12 16 23 34 36 45 56
93	6	7	12 15 23 34 36 45 56
94	6	6	12 13 23 34 35 36
95	6	6	13 23 34 35 45 56
96	6	6	12 23 25 34 35 56
97	6	6	12 13 23 34 35 56
98	6	6	12 23 35 34 45 56
99	6	6	12 23 24 45 46 56
100	6	6	12 23 34 45 46 56
101	6	6	12 14 23 34 35 36
102	6	6	12 14 23 25 34 36
103	6	6	12 23 42 35 45 56
103	6	6	12 14 23 34 35 56
105	6	6	12 15 23 34 45 46
106	6	6	12 16 23 34 45 56
107	6	5	16 26 36 46 56
108	6	5	14 24 34 45 56
109	6	5	13 23 34 45 46
110	6	5	12 23 34 36 45
111	6	5	12 23 34 45 46
112	6	5	12 23 34 45 56

Table 8: *6 node graphs labelled* 91 *to* 112 *in classification [1].*

9 Appendix B: sets $1_s,\ 2_s,\ 3_s,\ 4_s$ and 5_s

We give here the tables for the sets 1s, 2s, 3s, 4s and 5s for 5 node graph and 6 node graphs. The numbering of the graphs follow the ones given by Cvetkovic [1] for 5 and less nodes and 6 nodes graphs respectively.

9.1 1s

nodes	links	classification [1]	eigenvector	connection
3	2	3	$(-1, 0, 1)$	
4	3	8	$(0, 0, -1, 1)$	
4	4	6	$(1, 1, 0, -2)$	expansion on 5.3
5	4	28	$(1, 1, 0, -1, -1)$	
5	4	28	$(1, 0, 0, 0, -1)$	articulation on 5.3
5	4	28	$(1, 1, 1, 0, -3)$	star 4
5	4	29	$(0, 1, -1, 0, 0)$	articulation on 5.3
5	5	23	$(0, 0, 0, 1, -1)$	articulation on 5.3
5	5	23	$(1, 1, 0, -2, 0)$	expansion on 5.3
5	6	18	$(1, 1, 0, 1, -3)$	
5	6	20	$(1, 1, 0, -1, -1)$	articulation on 5.28
5	7	14	$(1, 1, 0, 1, -3)$	

Table 9: *Five node graphs with soft nodes and eigenvalue* 1.

nodes	links	classification [1]	eigenvector	connection
6	11	10	$(1,1,1,1,0,-4)$	link on 19
6	10	19	$(1,1,1,1,0,-4)$	link on 33
6	9	33	$(1,1,1,1,0,-4)$	link on 38
6	9	36	$(2,2,2,0,-3,-3)$	link on 61
6	9	38	$(1,1,1,1,0,-4)$	link 58
6	9	53	$(0,0,0,0,1,-1)$	link 75
6	9	53	$(1,1,1,0,-3,0)$	link 75
6	8	56	$(1,1,1,1,0,-4)$	expansion on 5.3
6	8	58	$(1,1,1,1,0,-4)$	expansion on 5.3
6	8	61	$(2,2,2,0,-3,-3)$	link on 94
6	7	75	$(0,0,0,0,1,-1)$	articulation on 5.3
6	7	75	$(1,1,0,1,-3,0)$	link on 101
6	7	78	$(0,0,0,0,1,-1)$	link on 101
6	7	79	$(1,1,0,0,0,-2)$	expansion on 5.3
6	7	79	$(1,1,0,-1,-1,0)$	link on 94
6	7	92	$(1,1,0,-1,-1,0)$	link on 106
6	6	94	$(1,1,0,-2,0,0)$	link on 107
6	6	94	$(3,3,0,-2,-2,-2)$	link on 107
6	6	94	$(1,-1,0,0,0,0)$	link on 95
6	6	95	$(1,-1,0,0,0,0)$	articulation on 5.3
6	6	97	$(1,1,0,-2,0,0)$	link on 108
6	6	99	$(1,0,-1,0,0,0)$	articulation on 5.3
6	6	101	$(0,0,0,0,1,-1)$	articulation on 5.3
6	6	106	$(0,1,1,0,-1,-1)$	link between two 5.3
6	5	107	$(-1,1,0,0,0,0)$	articulation on 5.3
6	5	107	$(1,1,-1,-1,0,0)$	soldering two 5.3 and articulation
6	5	107	$(1,1,0,-2,0,0)$	articulation on 5.3
6	5	108	$(-1,1,0,0,0,0)$	articulation on 5.3
6	5	108	$(-1,0,1,0,0,0)$	articulation on 5.3
6	5	109	$(-1,1,0,0,0,0)$	articulation on 5.3
6	5	109	$(0,0,0,0,-1,1)$	articulation on 5.3
6	5	111	$(0,0,0,0,-1,1)$	articulation on 5.3

Table 10: *Six node graphs with soft nodes and eigenvalue* 1.

9.2 2s

nodes	links	classification [1]	eigenvector	connection
4	5	5	$(1, 0, -1, 0)$	link on 5.7
4	4	7	$(1, 0, -1, 0)$	
4	4	7	$(0, 1, 0, -1)$	
5	8	12	$(1, 0, -2, 0, 1)$	link on 5.17
5	7	15	$(1, 0, -1, 0, 0)$	articulation on 5.7
5	7	15	$(1, 01, 0, -2)$	link on 5.17
5	7	17	$(1, 0, -2, 0, 1)$	
5	6	18	$(0, 1, 0, -1, 0)$	articulation 5.7
5	6	22	$(0, 1, 0, -1, 0)$	add a zero to 5.3 and articulation
5	6	22	$(1, 0, 0, -1, 0)$	add a zero to 5.3 and articulation
5	5	26	$(0, 1, 0, -1, 0)$	articulation 5.7

Table 11: *Five node graphs with soft nodes and eigenvalue* 2.

nodes	links	classification [1]	eigenvector	connection
6	12	5	$(1,1,0,-3,0,1)$	
6	11	11	$(1,1,1,0,-3,0)$	
6	11	13	$(1,0,-1,-1,0,1)$	
6	11	14	$(1,1,0,-3,0,1)$	
6	10	21	$(1,0,-1,-1,0,1)$	
6	10	21	$(0,0,1,-1,0,0)$	
6	10	29	$(1,1,1,0,-3,0)$	
6	10	31	$(1,0,-1,-1,0,1)$	link on 37
6	9	33	$(-2,0,1,1,0,0)$	link on 56
6	9	37	$(1,0,0,0,0,-1)$	link on 73
6	9	37	$(0,0,0,-1,0,1)$	link on 73
6	9	37	$(1,0,1,-1,0,-1)$	link on 73
6	9	40	$(0,0,0,0,1,-1)$	addition of a 0 to 5.3, articulation
6	9	49	$(0,0,1,-1,0,0)$	addition of a 0 to 5.3, articulation
6	9	49	$(1,0,-1,-1,0,1)$	link on 69
6	8	56	$(-1,1,0,0,0,0)$	addition of a 0 to 5.3, articulation
6	8	56	$(1,1,0,-2,0,0)$	link on 64
6	8	57	$(-1,0,1,0,0,0)$	link 93
6	8	61	$(0,0,0,0,-1,1)$	addition of a 0 to 5.3, articulation
6	8	64	$(1,1,0,-2,0,0)$	expansion of 5.7
6	8	66	$(0,0,1,0,-1,0)$	addition of a 0 to 5.3
6	8	69	$(0,1,1,-1,-1,0)$	link 5.7 and 5.3
6	8	71	$(0,0,0,1,-1,0)$	addition of a 0 to 5.3
6	8	73	$(1,0,0,0,0,-1)$	addition of a 0 to 5.3
6	8	73	$(0,0,0,-1,0,1)$	addition of a 0 to 5.3
6	8	73	$(1,0,1,-1,0,-1)$	soldering two 5.7
6	8	74	$(0,-1,0,1,0,0)$	link and articulation 5.7
6	7	75	$(0,-1,0,1,0,0)$	link 101
6	7	76	$(0,0,-1,0,1,0)$	link 103
6	7	81	$(0,-1,0,1,0,0)$	link and articulation 5.7
6	7	82	$(1,0,-1,0,-1,1)$	link 104
6	7	88	$(0,-1,0,1,0,0)$	articulation on 5.7
6	7	90	$(1,0,0,-1,0,0)$	articulation on 5.7
6	7	90	$(1,-1,0,0,0,0)$	articulation on 5.7
6	7	91	$(1,0,-1,0,0,0)$	addition of a 0 to 5.3
6	7	92	$(-1,1,1,1,-1,-1)$	link on 5.1
6	7	93	$(0,0,0,1,0,-1)$	addition of a 0 to 5.3, articulation
6	7	93	$(1,-1,-1,0,1,0)$	
6	6	101	$(0,1,0,-1,0,0)$	addition of a 0 to 5.3, articulation
6	6	103	$(0,0,1,-1,0,0)$	addition of a 0 to 5.3, articulation
6	6	104	$(0,1,0,-1,0,0)$	addition of a 0 to 5.3, articulation
6	6	104	$(1,0,-1,0,-1,1)$	articulation on 5.7

Table 12: *Six node graphs with soft nodes and eigenvalue* 2.

9.3 3s

nodes	links	classification [1]	eigenvector	connection
3	3	2	$(-1, 1, 0)$	
4	4	6	$(-1, 1, 0, 0)$	articulation 5.3
5	7	11	$(0, -1, 0, 0, 1)$	articulation 5.3
5	6	13	$(-1, 0, 0, -1, 0)$	articulation 5.3
5	8	13	$(0, 1, 0, 0, -1)$	articulation 5.3
5	7	16	$(-1, 1, 0, -1, 1)$	addition of zero to chain 4
5	7	17	$(0, 1, 0, -1, 0)$	articulation 5.3
5	6	20	$(-1, 1, 0, 0, 0)$	articulation 5.3
5	6	20	$(0, 0, 0, -1, 1)$	articulation 5.3
5	6	22	$(0, 0, -1, 0, 1)$	articulation 5.3
5	5	23	$(-1, 1, 0, 0, 0)$	articulation 5.3
5	5	25	$(-1, 1, 0, 0, 0)$	articulation 5.3

Table 13: *Five node graphs with soft nodes and eigenvalue* 3.

nodes	links	classification [1]	eigenvector	connection
6	13	3	$(-1,0,2,0,0,-1)$	link 8
6	12	6	$(0,-1,0,1,0,0)$	link 11
6	12	6	$(0,-1,0,-1,0,2)$	link 8
6	12	8	$(0,-2,0,0,1,1)$	link 11
6	11	11	$(1,0,-1,0,0,0)$	link 16
6	11	16	$(0,1,0,-1,0,0)$	link 25
6	11	16	$(0,-1,0,-1,0,2)$	link 17
6	11	17	$(0,2,0,-1,-1,0)$	link 52
6	11	17	$(1,0,-2,0,0,1)$	link 52
6	10	19	$(1,0,0,-1,0,0)$	link 39
6	10	25	$(0,0,0,-1,0,1)$	
6	10	29	$(1,0,-1,0,0,0)$	
6	10	32	$(0,1,0,-1,0,0)$	link 52
6	10	32	$(1,0,-1,0,0,0)$	link 52
6	9	36	$(0,0,0,0,1,-1)$	articulation 5.2
6	9	38	$(1,0,-1,0,0,0)$	articulation 5.13
6	9	38	$(0,1,0,-1,0,0)$	link 63
6	9	39	$(1,0,0,-1,0,0)$	link 63
6	9	51	$(1,1,0,-1,-1,0)$	link 106
6	9	51	$(0,0,1,-1,1,-1)$	link 106
6	9	52	$(0,1,0,1,0,-2)$	
6	9	52	$(1,0,1,0,-2,0)$	
6	9	52	$(1,0,-1,0,0,0)$	link 106
6	9	52	$(0,1,0,-1,0,0)$	link 106
6	8	58	$(-1,1,1,-1,0,0)$	link 79
6	8	61	$(0,-1,1,0,0,0)$	articulation 5.2
6	8	62	$(0,0,0,0,-1,1)$	articulation 5.2
6	8	63	$(0,1,0,-1,0,0)$	link 91
6	8	70	$(0,-1,1,1,-1,0)$	link 106
6	8	70	$(-1,0,1,1,0,-1)$	link 106
6	8	74	$(-1,0,1,-1,0,1)$	link 106
6	8	74	$(0,0,0,-1,0,1)$	link 106
6	8	74	$(-1,0,1,0,0,0)$	link 106
6	7	79	$(-1,1,0,0,0,0)$	articulation 5.2
6	7	79	$(0,0,0,-1,1,0)$	articulation 5.2
6	7	83	$(-1,1,0,0,0,0)$	articulation 5.2
6	7	84	$(-1,0,1,0,0,0)$	articulation 5.2
6	7	84	$(0,0,0,-1,0,1)$	articulation 5.2
6	7	88	$(0,0,0,0,-1,1)$	articulation 5.2
6	7	91	$(0,-1,0,1,0,0)$	
6	7	92	$(1,-1,0,1,-1,0)$	link 106
6	7	92	$(0,1,-1,0,1,-1)$	link 106
6	6	94	$(0,0,0,0,1,-1)$	articulation 5.2
6	6	97	$(0,0,0,0,1,-1)$	articulation 5.2
6	6	99	$(1,-2,1,2,-2,0)$	soldering P3 and C3
6	6	99	$(0,0,0,0,1,-1)$	articulation 5.2
6	6	100	$(1,-2,1,1,-1,0)$	soldering P3 and C3
6	6	100	$(0,0,0,0,1,-1)$	articulation 5.2
6	6	106	$(-1,0,1,-1,0,1)$	cycle 6
6	6	106	$(-1,1,0,-1,1,0)$	cycle 6

Table 14: *Six node graphs with soft nodes and eigenvalue* 3.

9.4 4s

nodes	links	classification [1]	eigenvector	connection
4	5	5	$(1, -2, 1, 0)$	
5	8	12	$(1, 0, 0, 0, -1)$	link on 17
5	7	14	$(0, 1, 0, -1, 0)$	
5	7	14	$(-1, 0, 0, 1, 0)$	link on 19
5	7	17	$(-1, 0, 0, 0, 1)$	
5	7	18	$(-2, 1, 0, 1, 0)$	articulation on 5
5	7	19	$(0, 1, 0, -1, 0)$	

Table 15: *Five node graphs with soft nodes and eigenvalue* 4.

nodes	links	classification [1]	eigenvector	connection
6	10	9	$(0,0,-1,1,0,0)$	
6	10	9	$(0,0,-1,1,0,0)$	
6	10	9	$(0,0,-1,1,0,0)$	
6	10	13	$(0,0,-1,1,0,0)$	
6	10	13	$(-1,0,0,0,0,1)$	
6	10	14	$(0,0,-1,0,1,0)$	
6	10	16	$(-1,0,1,0,0,0)$	
6	10	18	$(-1,0,-1,1,0,1)$	link 6.31
6	10	18	$(1,0,0,0,0,-1)$	link 6.31
6	10	21	$(1,0,0,0,0,-1)$	link 6.31
6	10	24	$(1,0,0,0,0,-1)$	link 6.31
6	10	29	$(0,0,0,1,0,-1)$	link 6.41
6	9	31	$(-1,0,-1,1,0,1)$	link 6.31
6	9	31	$(0,-0.6586,-0.2574,0.2574,0.6586,0)$	
6	9	31	$(1,0,0,0,0,-1)$	
6	9	33	$(0,0,-1,1,0,0)$	
6	9	35	$(0,-1,1,0,0,0)$	
6	9	36	$(0,-1,1,0,0,0)$	link 6.53
6	9	36	$(0,1,-1,0,0,0)$	link 6.53
6	9	41	$(0,0,0,1,0,-1)$	link 6.48
6	9	48	$(1,-1,1,0,-1,0)$	link 6.93
6	9	48	$(0,0,0,1,0,-1)$	link 6.49
6	9	49	$(0,-1,0,0,1,0)$	link 6.78
6	9	49	$(-1,0,0,0,0,-1)$	link 6.78
6	8	53	$(-1,1,0,0,0,0)$	link 6.78
6	8	53	$(0,1,-1,0,0,0)$	link 6.78
6	8	53	$(0,-1,1,0,0,0)$	articulation 5.5
6	8	55	$(-1,0,1,0,0,0)$	articulation 5.5
6	8	55	$(0,-1,1,0,0,0)$	articulation 5.5
6	8	61	$(0,0,0-2,1,1)$	
6	8	62	$(-1,1,0,0,0,0)$	link 6.78
6	8	64	$(-1,1,0,0,0,0)$	articulation 5.17
6	8	65	$(0,-1,1,0,0,0)$	
6	8	69	1 arbitrary zero	link 6.93
6	8	71	$(1,-1,1,0,-1,0)$	link 6.93
6	8	73	$(0,1,0,0,-1,0)$	soldering 5.7
6	7	75	$(-2,1,0,1,0,0)$	articulation 5.18
6	7	78	$(0,-1,0,1,0,0)$	articulation 5.5
6	7	80	$(0,0,-1,0,1,0)$	articulation 5.5
6	7	81	$(-2,1,0,1,0,0)$	articulation 5.18
6	7	82	$(0,-1,0,1,0,0)$	articulation 5.19
6	7	93	$(1,-1,1,0,-1,0)$	

Table 16: *Six node graphs with soft nodes and eigenvalue* 4.

9.5 5s

nodes	links	classification [1]	eigenvector	connection
5	10	10	$(1,-1,0,0,0)$	
5	10	10	$(0,1,-1,0,0)$	
5	10	10	$(0,0,1,-1,0)$	
5	10	10	$(0,0,0,1,-1)$	link on 5.11
5	9	11	$(1,0,-1,0,0)$	link on 5.12
5	9	11	$(0,0,1,-1,0)$	
5	8	12	$(0,1,0,-1,0)$	link 5.15
5	8	13	$(1,0,-2,1,0)$	add 2 soft nodes to 5.3
5	7	15	$(1,1,-3,1,0)$	add soft node to 4 star
5	7	15	$(0,0,-1,1,0)$	add 3 soft nodes to 5.1

Table 17: *Five node graphs with soft nodes and eigenvalue* 5.

nodes	links	classification [1]	eigenvector	connection
6	13	3	$(1,0,0,0,0,-1)$	
6	12	5	$(-1,1,0,0,0,0)$	link 6.12
6	12	5	$(1,1,0,0,0,-2)$	link 6.14
6	12	8	$(0,0,0,0,1,-1)$	link 6.12
6	12	8	$(1,0,0,-1,0,0)$	
6	11	10	$(1,-1,0,0,0,0)$	link 6.19
6	11	10	$(-1,1,-1,1,0,0)$	link 6.19
6	11	10	$(1,-1,-1,1,0,0)$	
6	11	11	$(1,1,0,0,0,-2)$	link 6.14
6	11	12	$(1,-1,0,0,0,0)$	link 6.14
6	11	14	$(1,1,0,0,0,-2)$	link 6.29
6	11	14	$(1,-1,0,0,0,0)$	
6	11	17	$(-1,0,0,1,-1,1)$	inserting a matching between two 5.2
6	11	17	$(0,0,0,1,-1,0)$	add 3 soft nodes to 5.1
6	10	19	$(-1,1,1,-1,0,0)$	link on 6.30
6	10	19	$(0,1,-1,0,0,0)$	link on 6.19
6	10	20	$(1,0,-2,1,0,0)$	link on 6.33
6	10	20	$(0,0,-1,1,0,0)$	link on 6.35
6	10	23	$(0,1,0,0,-1,0)$	link on 6.32
6	10	23	$(-1,0,-1,1,0,1)$	link on 6.30
6	10	26	$(0,0,0,1,0,-1)$	link on 6.32
6	10	29	$(1,-2,1,0,0,0)$	link on 6.32
6	10	30	$(1,-1,0,1,-1,0)$	link on 6.70
6	10	32	$(0,1,0,0,0,-1)$	link on 6.32
6	9	33	$(1,-3,1,1,0,0)$	link on 6.56
6	9	34	$(0,1,0,-1,0,0)$	link on 6.45
6	9	35	$(1,0,0,-1,0,0)$	link on 6.45
6	9	38	$(1,-1,1,-1,0,0)$	articulation on 5.13
6	9	39	$(1,-2,0,1,0,0)$	articulation on 5.13
6	9	44	$(-1,0,1,-1,0,1)$	link on 6.70
6	9	45	$(0,0,1,-1,0,0)$	link on 6.57
6	9	51	$(0,0,1,-1,1,-1)$	link on 6.70
6	8	56	$(1,1,-3,1,0,0)$	articulation on 5.15
6	8	57	$(0,-1,0,1,0,0)$	articulation on 5.15
6	8	70	$(-1,0,1,-1,0,1)$	insert matching on cycle 3

Table 18: *Six node graphs with soft nodes and eigenvalue* 5.

9.6 6s

nodes	links	classification [1]	eigenvector	connection
6	13	1	$(1,-1,0,0,0,0)$	
6	13	1	$(1,0,-1,0,0,0)$	
6	13	1	$(1,0,0,-1,0,0)$	
6	13	1	$(1,1,-2,0,0,0)$	link 6.3
6	13	2	$(1,-1,0,0,0,0)$	
6	13	2	$(1,0,-1,0,0,0)$	
6	13	2	$(1,0,0,-1,0,0)$	
6	13	3	$(1,-2,0,1,0,0)$	link 6.4
6	12	3	$(0,1,0,0,-1,0)$	link 6.4
6	12	4	$(1,-2,0,1,0,0)$	
6	12	4	$(0,1,0,0,-1,0)$	link 6.6
6	12	5	$(0,0,1,0,-1,0)$	link 6.7
6	12	6	$(1,1,0,0,0,-2)$	link 6.16
6	12	6	$(1,0,-1,0,0,0)$	link 6.7
6	12	7	$(0,1,0,0,-1,0)$	add four soft nodes to 5.1
6	12	8	1 arbitrarily placed 0	
6	11	9	$(1,1,0,-1,1,0)$	add two soft nodes to 5.7
6	11	11	$(1,1,1,-4,1,0)$	add a soft node to 5.28
6	11	13	1 arbitrarily placed 0	
6	11	16	$(1,0,1,0,-2,0)$	add 3 soft nodes to 5.2
6	9	37	$(1,0,0,-1,0,0)$	add 4 soft node to 5.1

Table 19: *Six node graphs with soft nodes and eigenvalue* 6.